MPA: A Multiprompt Physical Camouflage Attack Framework for Open-Vocabulary Object Detection in Remote Sensing Imagery

Ruiyang Jia, Fanjie Meng, Panyu Yue, Ruofei He, and Wei Sun

***Abstract*—Adversarial camouflage modifies the surface appearance of physical targets to mislead object detectors, making it a practical threat model for studying physically realizable attacks in remote sensing imagery. However, most existing approaches are designed for closed-set detectors with fixed category vocabularies and do not address the language-conditioned detection interface of open-vocabulary models. In open-vocabulary object detection, the same physical target can be queried using different textual descriptions, causing a camouflage pattern optimized for one prompt to remain detectable under another. This cross-prompt transfer uncertainty is further coupled with viewpoint variation in remote sensing imagery, where different target components become visible from different observation angles. In this article, we present a multiprompt physical camouflage attack framework for open-vocabulary object detectors (OVODs) in remote sensing imagery. The proposed method jointly optimizes adversarial camouflage across multiple semantically related prompts, suppressing prompt-conditioned detector responses rather than a single fixed class response. To ensure physical plausibility and visual concealment, we incorporate a controllable color constraint that enables scene-adaptive camouflage instantiation in natural environments such as forests and grasslands. In addition, we construct and release a multiview vehicle remote sensing dataset together with a reproducible data acquisition pipeline, enabling systematic evaluation under diverse viewpoints and observation conditions. Experimental results demonstrate that the proposed method achieves high attack success rates (ASRs) across multiple prompts, viewpoints, and observation conditions while maintaining strong visual stealth. These results provide a reproducible testbed for studying physical vulnerabilities of open-vocabulary remote sensing perception systems. Code and dataset are available at https://github.com/jryxxx/MPAttack**

***Index Terms*—Adversarial camouflage, aerial imagery, multiprompt adversarial attack, open-vocabulary object detection.**

I. Introduction

With the rapid advancement of deep neural networks (DNNs), these models have achieved remarkable success in a wide range of vision tasks, including remote sensing and aerial image analysis [1], [2], [3], [4], [5]. Despite their strong detection performance, DNN-based detectors are still vulnerable to adversarial examples [6], which may lead to missed or spurious detections in practical scenarios such as autonomous driving, surveillance, and remote sensing interpretation. This article investigates physical adversarial attacks against open-vocabulary object detectors (OVODs) in remote sensing imagery, a setting where the detector's semantic interface is specified by text queries at inference time. This capability is useful for remote sensing interpretation because target vocabularies can change with the task, scene, and mission requirements. It also creates a security problem that closed-set physical attacks do not address: a physical camouflage pattern must remain effective when the same object is queried using different textual descriptions, while also preserving robustness to viewpoint variation and scale changes.

Adversarial examples are typically divided into digital and physical categories. Digital adversarial examples perturb pixel-level inputs in the digital domain [7], [8], whereas physical adversarial examples modify the appearance of objects or their surroundings in the physical environment to mislead detectors [9], [10]. Compared with digital attacks, physical adversarial attacks are more realistic and pose greater risks in real-world scenarios, since they do not require direct access to sensor data. However, designing effective physical attacks remains highly challenging [11], as the perturbations must preserve effectiveness under diverse viewing conditions, illumination changes, scale variations, and sensor noise.

Adversarial camouflage methods for object detection typically aim to preserve attack effectiveness across wide viewing-angle ranges by modifying the target object's surface appearance. Prior work often constructs full-coverage camouflage patterns, and the state-of-the-art leverages differentiable neural rendering to connect 3-D surface textures with 2-D image observations, enabling end-to-end optimization under physically realistic imaging conditions [12], [13]. Broadly, existing camouflage strategies fall into two categories: one optimizes a universal pattern that is tiled or repeated over the object surface, often relying on physical modeling or nondifferentiable rendering approximations [14], [15]; the other adopts differentiable neural rendering to directly optimize 3-D textures, yielding greater flexibility and robustness to viewpoint variation [12], [16].

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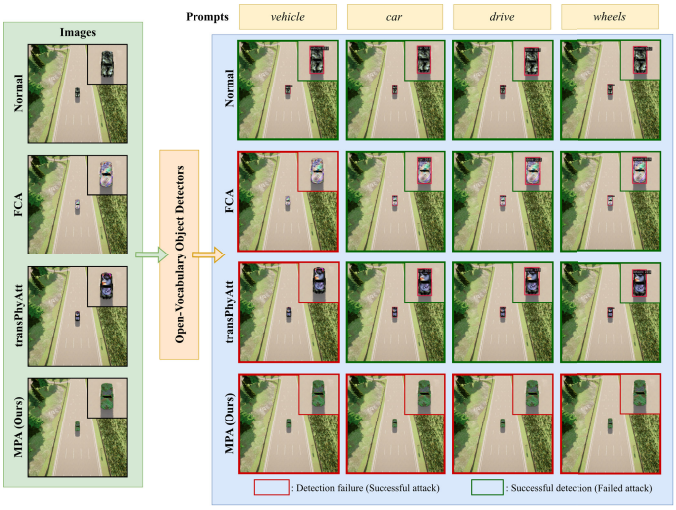


Fig. 1. Detection results under different semantic prompts in open-vocabulary object detection. Single-prompt physical attacks (e.g., FCA and transPhyAtt) fail when the textual query changes, whereas the proposed multiprompt attack (MPA) remains effective across all evaluated prompts (*vehicle*, *car*, *drive*, and *wheels*).

Despite these advances, existing physical camouflage attacks remain insufficient for open-vocabulary remote sensing detection. Most aerial attacks are patch-based [17], [18], which limits deployability and robustness across observation conditions, while the few coating-style approaches [19] are typically optimized for specific viewpoints. More importantly, prior physical attacks are mainly formulated for closed-set detectors with a fixed category vocabulary. In that setting, the attacker suppresses one predetermined class response. OVODs introduce a different attack interface: categories are specified by textual queries at inference time, and detection scores are produced through vision–language alignment [20], [21]. The same physical target can therefore be queried by category names, functional descriptions, or structural terms, and these prompts may rely on different visual evidence. We refer to the resulting uncertainty as cross-prompt transfer uncertainty. Fig. 1 illustrates this effect: single-prompt physical attacks lose effectiveness when the textual query changes, even though the image content is unchanged. In remote sensing imagery, this semantic uncertainty is further coupled with viewpoint variation because changing azimuth and elevation alters which target components are visible. These factors motivate a physical camouflage framework that suppresses prompt-conditioned detector responses across both semantic queries and observation viewpoints while maintaining visual concealment.

We propose a multiprompt, physically grounded camouflage framework for attacking OVODs in remote sensing imagery. The framework treats adversarial camouflage as a texture-optimization problem on 3-D vehicle surfaces, using a differentiable neural renderer to project optimized textures into image space under physically realistic observation conditions. To address the language-conditioned attack surface of OVODs, the attack target is defined as a set of semantically related prompts rather than a single class label. The multiprompt loss evaluates each textual query independently and jointly suppresses the corresponding detector responses, reducing overfitting to one prompt and improving robustness under alternative descriptions of the same object. A controllable color constraint keeps the generated textures within a scene-appropriate palette, reducing visual conspicuity without sacrificing attack effectiveness. We evaluate the approach across different prompts, detectors, viewpoints, and observation conditions. We also construct and release two multiview vehicle datasets, an armored vehicle dataset in a forest scene and a civilian vehicle dataset in an urban scene, together with a reproducible data acquisition pipeline, to benchmark camouflage generation and evaluate cross-target transferability.

The main contributions of this work are summarized as follows.

1) We formulate physically realizable adversarial camouflage against OVODs as a language-conditioned physical attack problem. Unlike closed-set attacks that suppress a fixed class response, the attack objective is defined over a set of textual queries that may describe the same target through category, function, or structure. This formulation identifies cross-prompt transfer uncertainty as a core challenge for physical attacks against OVODs in remote sensing imagery.
2) We construct a reproducible multiview remote sensing dataset and release a complete data acquisition pipeline for evaluating adversarial camouflage under diverse environments, target categories, and observation viewpoints. The curated dataset serves as a benchmark for studying robustness and transferability of physical attacks in open-vocabulary detection scenarios and enables scalable extension to new targets and scenes.
3) We develop a problem-driven multiprompt adversarial optimization framework that jointly optimizes adversarial camouflage over semantically related textual queries. By integrating a specialized multiprompt loss with perceptual and color regularization, the proposed method achieves robust attack performance across prompt variations and viewing angles while maintaining low visual conspicuity and physical plausibility.

II. Related Work

In this section, we first review the evolution of object detection techniques and then summarize physical adversarial attack methods relevant to detection.

A. Object Detection

Traditional object detection is commonly formulated as a closed-set recognition problem, where training and evaluation share a predefined category vocabulary, such as those in COCO [22] and Objects365 [23]. Representative architectures include region-based detectors [24], [25], dense prediction methods [26], [27], and query-based Transformer detectors [28], [29]. These methods have substantially improved localization accuracy, multiscale representation, and end-to-end detection in natural-image benchmarks.

Remote sensing object detection introduces additional challenges beyond natural-image detection. Targets are often

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represented by only a few pixels, densely distributed across large scenes, observed at substantially different scales, and captured with arbitrary orientations. Recent aerial small-object detection methods therefore emphasize cross-scale interaction and hierarchical feature aggregation. Zhang and Liao [30] combine a cross-scale adaptive Transformer with hierarchical feature fusion to strengthen semantic representations while preserving the spatial details required for small-object localization.

Open-vocabulary object detection relaxes the fixed-vocabulary assumption by conditioning detection on textual category descriptions. Early methods adopted base-to-novel evaluation protocols and transferred semantic knowledge from pretrained vision–language models [31], [32], [33]. Subsequent studies improved regional alignment, large-scale pretraining, and text-conditioned localization [34], [35], [36]. Representative general-purpose detectors include GLIP [37], which learns phrase-aware region representations; GroundingDINO [21], which integrates grounded pretraining with cross-modal feature fusion; and YOLO-World [38], which incorporates region-text alignment into a real-time one-stage detector.

Recent studies further adapt open-vocabulary detection to aerial and remote sensing imagery. CastDet [39] employs a CLIP-activated student-teacher architecture, uses RemoteCLIP as an auxiliary teacher, and maintains a dynamic label queue to improve class-agnostic proposals and pseudo-label quality for novel aerial categories. LAE-DINO [40] constructs the LAE-1M remote sensing detection dataset and introduces dynamic vocabulary construction and visual-guided text prompt learning to reduce the domain gap between natural and remote sensing imagery. OpenRSD [41] develops a universal open-prompt framework that supports multimodal prompts and both horizontal and oriented object detection.

Beyond accuracy-oriented detection, efficient remote sensing perception is also important for resource-constrained platforms. Spiking neural networks provide one promising direction because of their event-driven computation. Saliency-guided spiking models have been applied to target detection in high-resolution remote sensing images [42]. RepSViT [43] combines spiking computation with vision Transformer components for efficient object recognition in satellite on-orbit applications. SNN-based methods have also been investigated for event-driven optical flow estimation [44].

B. Physical Adversarial Attack

Physical adversarial attacks for object detection are commonly grouped into patch-based and camouflage-based methods, according to how perturbations are instantiated.

Patch-based attacks generate universal adversarial patches that are designed to remain effective under a range of geometric and photometric transformations [45], [46]. Early work such as that of Xie et al. [47] uses a surrogate rendering-to-detection pipeline to iteratively optimize patches in a white-box setting, while subsequent methods [48] explore discrete or query-based search strategies to produce deployable patches that can be tiled or scaled on object surfaces. Although these methods can yield strong results in controlled environments, their real-world effectiveness is often limited by insufficient physical realism and constrained generalization.

Camouflage-based attacks modify object geometry or surface appearance (textures) to achieve more pervasive and physically plausible concealment. Recent advances exploit differentiable neural rendering to bridge 3-D surface representations and 2-D image observations, enabling end-to-end optimization of texture (and in some cases shape) under simulated imaging conditions [12], [16], [49], [50]. Such approaches typically produce full-coverage or coating-style adversarial patterns that better withstand viewpoint variation compared with local patches, but they still face challenges in maintaining visual naturalness and semantic generalization—especially when evaluated under open-vocabulary detection settings.

C. Research Gaps and Motivation

Although prior work has advanced both patch-based and camouflage-based physical attacks, three limitations remain under open-vocabulary remote sensing detection. First, many aerial attacks rely on localized patches or viewpoint-specific coatings, which limits physical coverage and stability under changing observation geometry [17], [18], [19]. Second, existing attack objectives are largely closed-set: they optimize against fixed detector classes and do not model the prompt-conditioned responses produced by vision–language detectors. Third, physical plausibility and visual concealment are often treated as secondary constraints, causing effective adversarial patterns to deviate from natural scene statistics. These limitations motivate a framework that jointly addresses physical robustness, cross-prompt stability, and scene-compatible camouflage generation for OVODs in remote sensing imagery.

III. Method

This section describes the proposed method. We first define the overall framework, then detail the color constraint, the loss design, and the optimization procedure.

A. Overall Framework

Fig. 2 presents the overall pipeline of the proposed MPA framework. The method starts from a textured 3-D vehicle model and a set of remote sensing observation configurations, including camera poses, vehicle poses, foreground masks, and target bounding boxes. A learnable adversarial texture is first mapped to a physically plausible color range and then projected onto the vehicle surface through differentiable neural rendering. The rendered vehicle is composited with the background image to form an adversarial training sample. The resulting image is queried by an open-vocabulary detector using multiple semantically related prompts, and the prompt-conditioned detection responses associated with the target region are aggregated into an MPA loss. Gradients are then backpropagated through the detector, image composition, color constraint, and renderer to update the surface texture. Through this loop, the optimized camouflage is encouraged to suppress detections across both viewpoint variation and alternative textual descriptions of the same target.

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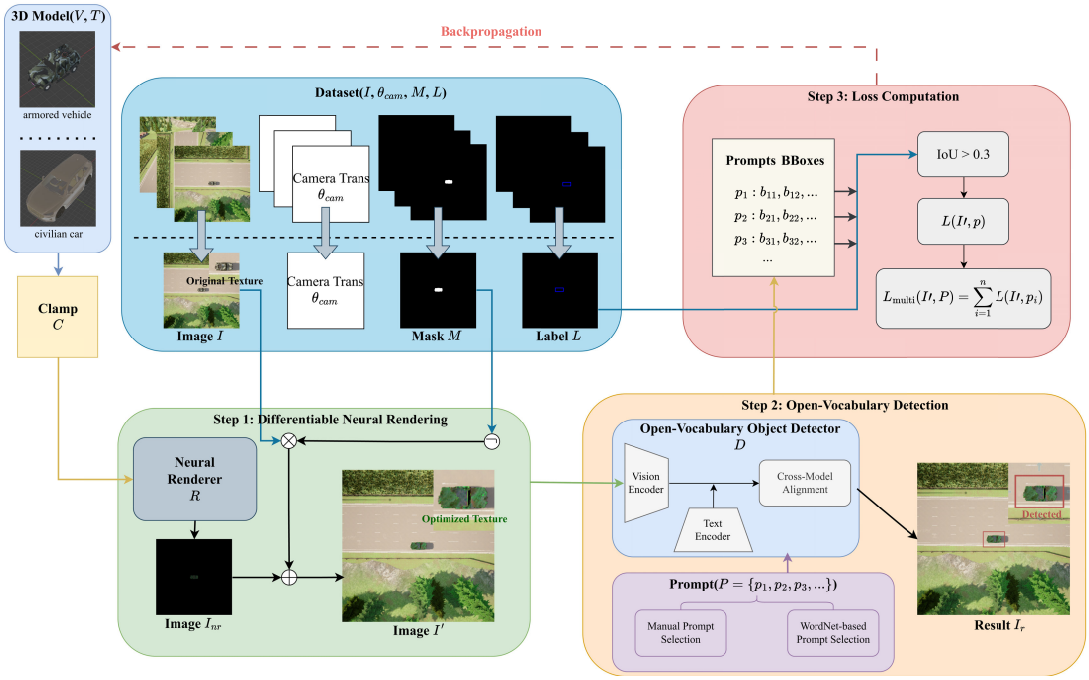


Fig. 2. Overview of the proposed MPA framework. Ground-level images with vehicle and camera poses, instance masks, and annotations are collected from a photorealistic simulator. Adversarial textures are rendered onto vehicle surfaces, with color constraints applied to reduce visual saliency. The camouflaged vehicles are then embedded into diverse scenes, and the textures are optimized via backpropagation with a custom loss.

The target vehicle is represented as a triangular mesh $V = (\mathbf{X}, \mathbf{F})$, where $\mathbf{X} \in \mathbb{R}^{N_v \times 3}$ denotes the 3-D vertex coordinates and $\mathbf{F} \in \mathbb{N}^{N_f \times 3}$ denotes the triangular face indices. The original texture T and the learnable adversarial texture T' are defined in the 2-D UV texture domain. The training set contains N observation tuples

$$\mathcal{D} = \left\{\left(I_k, \theta_{\text{cam},k}, M_k, L_k\right)\right\}_{k=1}^{N} \tag{1}$$

where $I_k \in [0,1]^{H \times W \times 3}$ is an RGB image, $\theta_{\text{cam},k} \in \mathbb{R}^{d_\theta}$ contains the camera and vehicle pose parameters, $M_k \in \{0,1\}^{H \times W}$ is the vehicle foreground mask, and $L_k \in \mathbb{R}^4$ is the ground-truth bounding box.

The differentiable renderer is formulated as

$$R : \mathcal{V} \times [0,1]^{H_T \times W_T \times 3} \times \mathbb{R}^{d_\theta} \to [0,1]^{H \times W \times 3} \tag{2}$$

where $\mathcal{V}$ denotes the space of triangular vehicle meshes. Given the mesh V, the color-constrained texture $C(T')$, and the pose parameters θ_{cam}, the rendered vehicle image is

$$I_{\text{nr}} = R\left(V, C\left(T'\right), \theta_{\text{cam}}\right) \tag{3}$$

with $I_{\text{nr}} \in [0,1]^{H \times W \times 3}$.

The binary vehicle mask M is expanded along the channel dimension to $\widetilde{M} \in \{0,1\}^{H \times W \times 3}$. The final adversarial image is obtained by compositing the rendered vehicle foreground with the original background

$$I' = \widetilde{M} \odot I_{\text{nr}} + \left(\mathbf{1} - \widetilde{M}\right) \odot I \tag{4}$$

where $\odot$ denotes element-wise multiplication and $I' \in [0,1]^{H \times W \times 3}$.

Let $\mathcal{S}$ denote the textual prompt space. An open-vocabulary detector is defined as a prompt-conditioned mapping

$$D : [0,1]^{H \times W \times 3} \times \mathcal{S} \to 2^{\mathbb{R}^4 \times [0,1]} \tag{5}$$

where

$$D\left(I', p\right) = \left\{\left(b_j, s_j\right)\right\}_{j=1}^{m_p} \tag{6}$$

is the prediction set for prompt $p \in \mathcal{S}$. Each $b_j \in \mathbb{R}^4$ is a predicted bounding box, $s_j \in [0,1]$ is the corresponding confidence score, and m_p is the number of predictions.

The attack target is defined over a prompt set

$$P = \{p_i\}_{i=1}^{n} \subset \mathcal{S} \tag{7}$$

where $n = |P|$. The adversarial texture is optimized by aggregating the losses induced by all prompts

$$T'^{*} = \underset{T' \in [0,1]^{H_T \times W_T \times 3}}{\arg\min} \sum_{i=1}^{n} \mathcal{L}\left(I', p_i; L\right) \tag{8}$$

where $L \in \mathbb{R}^4$ is the ground-truth bounding box and $\mathcal{L}(I', p_i; L) \in \mathbb{R}_{\geq 0}$ is the attack loss associated with prompt p_i.

B. Color Constraint

To reduce the visual conspicuity of the generated camouflage, the adversarial texture is mapped to a configurable RGB color interval. Let $T' \in [0,1]^{H_T \times W_T \times 3}$ denote the learnable texture parameters, and let $c_{\min}, c_{\max} \in [0,1]^3$ denote the channel-wise lower and upper color bounds, with $c_{\min} \preceq c_{\max}$. The color constraint is

$$C\left(T'\right) = T' \odot \left(c_{\max} - c_{\min}\right) + c_{\min} \tag{9}$$

applied element-wise, so that $C(T') \in [c_{\min}, c_{\max}]^{H_T \times W_T \times 3}$. The interval can be configured according to the deployment environment, including vegetation-dominated, desert, and urban scenes.

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The bounds $c_{\min}$ and $c_{\max}$ define the allowable color range of the optimized camouflage. They can be specified according to a desired camouflage palette or derived from scene statistics before optimization. Once selected, the interval is fixed during texture optimization and applied uniformly to all rendered training samples. This design separates the color prior from the attack objective, allowing the same optimization framework to be evaluated under different color-range choices.

C. Loss Design

The attack loss is designed to suppress detector responses associated with the target under multiple textual queries.

1) IoU-Based Confidence Loss: For an adversarial image $I' \in [0,1]^{H\times W\times 3}$ and a prompt $p \in \mathcal{S}$, the detector outputs

$$D\left(I', p\right) = \left\{\left(b_j, s_j\right)\right\}_{j=1}^{m_p} \tag{10}$$

where $b_j \in \mathbb{R}^4$ and $s_j \in [0,1]$. We first retain predictions whose confidence scores exceed a threshold

$$D_{\text{conf}}\left(I', p\right) = \left\{\left(b_j, s_j\right) \in D\left(I', p\right) \mid s_j > \tau_{\text{conf}}\right\}. \tag{11}$$

We then retain predictions that spatially overlap with the target box $L \in \mathbb{R}^4$

$$D_{\text{IoU}}\left(I', p; L\right) = \left\{\left(b_j, s_j\right) \in D_{\text{conf}}\left(I', p\right) \mid \text{IoU}\left(b_j, L\right) > \tau_{\text{IoU}}^{\text{train}}\right\}. \tag{12}$$

We use $\tau_{\text{conf}} = 0.1$ and set the default training IoU threshold to $\tau_{\text{IoU}}^{\text{train}} = 0.3$. The confidence threshold removes weak detector responses, while the IoU threshold restricts the loss to predictions associated with the target region. This threshold is analyzed in the ablation study by jointly varying the training and evaluation IoU thresholds.

Let $N_p = |D_{\text{IoU}}(I', p; L)|$ be the number of retained predictions for prompt p. The single-prompt attack loss is defined as

$$\mathcal{L}\left(I', p; L\right) = \begin{cases} \dfrac{1}{N_p} \displaystyle\sum_{\left(b_j, s_j\right) \in D_{\text{IoU}}(I', p; L)} s_j, & N_p > 0 \\ 0, & N_p = 0. \end{cases} \tag{13}$$

Minimizing this loss suppresses the confidence scores of target-associated predictions that satisfy both the confidence and localization criteria.

2) Multiprompt Joint Attack Loss: Let $P = \{p_i\}_{i=1}^n \subset \mathcal{S}$ denote a set of semantically related prompts describing the target. Each prompt p_i is processed independently by the detector, and the corresponding single-prompt loss $\mathcal{L}(I', p_i; L)$ is computed from $D(I', p_i)$. The MPA loss is the direct sum of these prompt-conditioned losses

$$\mathcal{L}_{\text{multi}}\left(I', P; L\right) = \sum_{i=1}^{n} \mathcal{L}\left(I', p_i; L\right) \tag{14}$$

where $\mathcal{L}_{\text{multi}}(I', P; L) \in \mathbb{R}_{\geq 0}$.

Open-vocabulary detectors may support evaluating multiple textual queries or category phrases within a single inference call. While this is efficient for standard detection inference, adversarial texture optimization imposes a different requirement: the loss associated with each prompt should reflect the detector response induced by that prompt under an independent query condition. When semantically related prompts are evaluated jointly, their confidence scores may be affected by score competition or cross-prompt suppression within the detector output. Because the proposed attack objective directly minimizes target-associated confidence scores, such changes can alter the loss magnitude and weaken the gradient signal used for texture optimization. MPA therefore computes each prompt-conditioned loss through an independent detector call and sums the resulting losses. This strategy is used consistently for GLIP, GroundingDINO, and YOLO-World, so that each textual query contributes its own prompt-conditioned response to the multiprompt objective.

This formulation is motivated by the interaction between semantic prompts and viewpoint-dependent visual evidence. Under aerial observation, changes in azimuth and elevation expose different target components, such as the vehicle body, wheels, and other structural cues. Different prompts may therefore activate different detector responses for the same physical target. Optimizing the summed loss over P penalizes these prompt-conditioned responses simultaneously, which reduces dependence on a single textual query. The empirical analysis in Section IV-C examines this behavior across 36 viewpoints.

Algorithm 1 MPA

Input: Training set $\mathcal{D} = \{(I_k, \theta_{\text{cam},k}, M_k, L_k)\}_{k=1}^N$, prompt set $P = \{p_i\}_{i=1}^n$, total epochs E, color constraint C, 3D vehicle mesh V, differentiable renderer R, object detector D

Output: Adversarial texture $C(T'^*)$

```
Initialize T' ∈ [0,1]^{H_T × W_T × 3} with random values
for e = 1 to E do
    for each (I, θ_cam, M, L) ∈ D do
        T̂ ← C(T')
        I_nr ← R(V, T̂, θ_cam)
        Expand M to M̃ ∈ {0,1}^{H×W×3}
        I' ← M̃ ⊙ I_nr + (1 − M̃) ⊙ I
        L_multi ← 0
        for i = 1 to n do
            Y_i ← D(I', p_i)
            Compute L(I', p_i; L) from Y_i
            L_multi ← L_multi + L(I', p_i; L)
        end for
        Update T' by backpropagating L_multi
    end for
end for
return C(T'^*)
```

D. Optimization Process

The adversarial texture is optimized by minimizing the MPA loss. For each training sample, the learnable texture is first mapped to the selected color interval and rendered onto the 3-D vehicle mesh. The rendered vehicle foreground is then composited with the background image. The detector processes each prompt independently, and the resulting

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TABLE I
COMPOSITION OF THE MULTIVIEW VEHICLE BENCHMARK. EACH TARGET TYPE IS REPRESENTED BY ONE 3-D MODEL, AND ALL IMAGES ARE RENDERED AT 800 × 800 PIXELS

Target Type	Scene	Models	Positions	Views/Position	Generated	Removed	Retained	Train	Test
Armored vehicle	Forest	1	200	48	9,600	0	9,600	7,680	1,920
Civilian vehicle	Urban	1	200	48	9,600	97	9,503	7,602	1,901
Total	Two scenes	2	400	48	19,200	97	19,103	15,282	3,821

prompt-conditioned losses are accumulated. Gradients are backpropagated through the detector, image composition operation, color constraint, and differentiable renderer to update the texture parameters. Algorithm 1 summarizes the optimization procedure.

IV. EXPERIMENTS

In this section, we first outline the experimental settings and subsequently demonstrate the effectiveness of the proposed attack framework through extensive evaluations in both digital and physical environments.

A. Experimental Setup

1) Dataset: We construct a multiview vehicle benchmark in a simulated aerial observation environment for controlled evaluation of physically realizable adversarial camouflage. The benchmark is used for attack generation and robustness evaluation. All evaluated detectors use publicly released pretrained weights and remain frozen during camouflage optimization. The benchmark is not used to train or fine-tune any detector.

The benchmark contains two vehicle target types, each represented by one 3-D model. The armored vehicle is rendered in a forest scene, and the civilian vehicle is rendered in an urban scene. These two target types provide controlled variation in vehicle geometry, surface layout, and background context while preserving a reproducible physical attack setting.

For each target model, 200 vehicle positions are sampled along road trajectories. At every position, 48 observations are rendered from 12 azimuth angles at 30° intervals and four elevation angles. Each sample contains an RGB image, the vehicle mask, camera and vehicle poses, and a bounding-box annotation. All images are rendered at 800 × 800 pixels.

Table I summarizes the benchmark composition. The initial sampling produces 9600 images for each target type. All armored-vehicle images are retained and split into 7680 training images and 1920 test images. For the civilian vehicle, 97 images in which the target is fully occluded are removed. The remaining 9503 images are split into 7602 training images and 1901 test images.

In this work, open-vocabulary refers to the attacked detector interface and evaluation protocol rather than the number of physical target categories in the benchmark. GLIP, GroundingDINO, and YOLO-World are open-vocabulary detectors because their target categories are specified through textual queries at inference time instead of being restricted to a fixed detector-specific label set. During evaluation, the same physical target is queried using semantically related textual prompts selected for the evaluated detector and experiment. An attack is counted as successful only when the target remains undetected under every prompt in the evaluation set. The benchmark therefore provides controlled physical targets and scene configurations for evaluating query-conditioned detection robustness, while the open-vocabulary property comes from the detector interface and the cross-prompt evaluation protocol.

2) Comparison Methods: Most existing physical adversarial attack methods are originally designed for closed-set object detectors and optimize camouflage against a fixed class label. To establish meaningful baselines for the open-vocabulary setting, the experimental evaluation distinguishes between victim detectors and attack methods. GLIP, GroundingDINO, and YOLO-World are fixed open-vocabulary victim detectors used to evaluate robustness, whereas FCA, transPhyAtt, and MPA are the physical attack methods under comparison.

FCA [12] and transPhyAtt [51] are representative camouflage attacks originally designed for closed-set detection. We reimplement and adapt them using the same neural rendering pipeline and detector interface as MPA. For each adapted baseline, adversarial camouflage is optimized using a single textual prompt, which provides a unified single-prompt attack formulation while preserving the core design of the original method.

MPA uses the same experimental setting but replaces the single-prompt objective with a multiprompt objective that jointly suppresses prompt-conditioned detector responses. This design directly targets cross-prompt transfer uncertainty, which is not modeled by single-prompt physical attack baselines. Compared with adapted closed-set physical attacks, MPA has three advantages in the OVOD setting. It optimizes against multiple textual queries simultaneously, evaluates success under a stricter all-prompts criterion, and constrains the texture color range to improve physical plausibility and visual concealment. These properties are aligned with the language-conditioned detection interface of OVODs and the multiview observation characteristics of remote sensing imagery.

We conduct systematic comparisons between FCA, transPhyAtt, and MPA across multiple open-vocabulary victim detectors. To ensure fairness, all attack methods are evaluated under the same conditions, including the same renderer, data split, prompt configuration, hyperparameter settings, training epochs, optimizer, and evaluation thresholds.

3) Evaluation Metrics: Our objective is to induce false negatives by concealing the target vehicle from the detector. Accordingly, we adopt the attack success rate (ASR) [52] as the primary evaluation metric, defined as the percentage of evaluated target instances that become undetected or misdetected after applying adversarial perturbations.

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Algorithm 2 MP-ASR

Input: Test set $\mathcal{D}_{\text{test}} = \{(I, L)\}$, attack prompts P, object detector D, confidence threshold c_t
Output: MP-ASR r

Initialize the number of evaluated instances: $n_c \leftarrow |\mathcal{D}_{\text{test}}|$
Initialize the number of adversarial detections: $n_a \leftarrow 0$
for (I, L) in $\mathcal{D}_{\text{test}}$ **do**
  Initialize adversarial detection flag: $r_a \leftarrow$ False
  Generate adversarial image I' using Eq. 4
  for p in P **do**
    $D_{\text{conf}} = \{(b, s) \in D(I', p) \mid s > c_t\}$
    $D_{\text{IoU}} = \{(b, s) \in D_{\text{conf}} \mid \text{IoU}(b, L) > 0.3\}$
    if $D_{\text{IoU}} \neq \emptyset$ **then**
      $r_a \leftarrow$ True
    end if
  end for
  if $r_a =$ True **then**
    $n_a \leftarrow n_a + 1$
  end if
end for
return $r = (n_c - n_a)/n_c$

To further assess the effectiveness of MPAs, we introduce the MPA success rate (MP-ASR), defined as follows: for a given target instance, if it remains undetected under all input prompts, the attack is deemed successful; otherwise, if any prompt leads to a correct detection, the attack is considered a failure.

During evaluation, we distinguish between missed detections and false detections using the confidence score and the intersection-over-union (IoU) with the ground-truth box. A prediction is counted as a valid target detection only when its confidence score is above the evaluation confidence threshold and its IoU with the ground-truth box is above the evaluation IoU threshold $\tau_{\text{IoU}}^{\text{eval}}$. Otherwise, the target is treated as missed or misdetected for the corresponding prompt. In the main experiments, we report MP-ASR under confidence thresholds from 0.1 to 0.5 and use $\tau_{\text{IoU}}^{\text{eval}} = 0.3$ by default. The sensitivity to $\tau_{\text{IoU}}^{\text{eval}}$ is analyzed in Section IV-I3. The detailed computation of MP-ASR is summarized in Algorithm 2.

4) Experimental Details: We conduct white-box attack experiments on three widely used open-vocabulary victim detectors: GLIP [37], GroundingDINO [21], and YOLO-World [38]. These detectors are selected to provide a controlled and reproducible evaluation across complementary OVOD paradigms. GLIP performs phrase grounding through region-phrase alignment, GroundingDINO uses Transformer-based grounded detection with cross-modal feature fusion, and YOLO-World incorporates region-text alignment into a real-time one-stage detector. All three use publicly released pretrained weights and remain frozen during camouflage optimization.

We use these general-purpose OVODs because they provide stable public implementations and clear prompt-conditioned detection interfaces, which are necessary for integrating differentiable rendering, detector inference, and gradient-based texture optimization into a reproducible attack pipeline. GLIP and GroundingDINO are supported within the MMDetection [53] ecosystem, while YOLO-World provides an official implementation. Recent remote sensing-specific OVOD variants often introduce additional domain adaptation modules, dataset-specific vocabularies, or customized inference pipelines, and adapting them to a differentiable physical attack loop requires separate engineering effort. Evaluating MPA on such detectors is therefore left as future work, as discussed in Section VI.

All experiments are performed on an NVIDIA RTX 4090 GPU with 24 GB of memory. The number of optimization iterations is set to 5, and the Adam optimizer is adopted with a learning rate of 0.01.

B. Digital Adversarial Attack

In this section, we evaluate the proposed method in digital settings. FCA and transPhyAtt are optimized using a single textual prompt (*vehicle*), while MPA is optimized using four semantically related prompts: *vehicle*, *car*, *drive*, and *wheels*. All methods use the same data, rendering pipeline, and evaluation protocol.

Table II reports MP-ASR at confidence thresholds from 0.1 to 0.5, together with the average across these thresholds. The reported values are absolute MP-ASR scores at each operating point. Since lower confidence thresholds count more detector predictions as valid detections, they define stricter attack conditions. MPA remains effective under these strict settings, achieving 94.90%, 98.85%, and 99.58% MP-ASR at the lowest threshold of 0.1 on GroundingDINO, GLIP, and YOLO-World, respectively. Compared with the strongest baseline for each detector, MPA improves Avg. MP-ASR from 53.11% to 98.88% on GroundingDINO, from 97.03% to 99.57% on GLIP, and from 88.86% to 99.90% on YOLO-World. These results show that the improvement is consistent across thresholds rather than being limited to high-threshold cases.

The ablation study in Section IV-I7 further isolates the effect of multiprompt optimization. The single-prompt augmentation baseline uses the same prompt pool but optimizes only the strongest prompt response at each step, whereas MPA jointly optimizes all prompt-conditioned losses. Under the same training and evaluation protocol, multiprompt optimization improves Avg. MP-ASR from 86.48% to 98.81% on GroundingDINO, from 86.30% to 98.70% on GLIP, and from 91.64% to 99.93% on YOLO-World. This comparison indicates that the improvement mainly comes from jointly suppressing prompt-conditioned responses under the all-prompts MP-ASR criterion.

Fig. 3 presents a visual illustration of the adversarial camouflage and its attack effectiveness under different textual prompts. Both the training and testing of the adversarial textures are conducted on the GroundingDINO model. The naturally camouflaged vehicle (normal) fails to achieve effective attacks under any prompt. Due to being optimized for a single prompt only, FCA and transPhyAtt are effective under one specific prompt but fail under others. In contrast, our method (MPA) successfully performs attacks under all prompts

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TABLE II
MP-ASR EVALUATION ON THREE OPEN-VOCABULARY DETECTORS WITH FOUR PROMPTS

Model	Method	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
GroundingDINO	Normal	0.00%	0.00%	0.36%	6.46%	30.94%	7.55%
	FCA	0.00%	0.26%	16.67%	73.54%	97.92%	37.68%
	transPhyAtt	0.36%	26.20%	59.11%	84.64%	95.26%	53.11%
	MPA (Ours)	**94.90%**	**99.69%**	**99.84%**	**99.95%**	**100.00%**	**98.88%**
GLIP	Normal	3.54%	4.38%	15.73%	35.73%	62.81%	24.44%
	FCA	90.57%	94.84%	99.84%	99.95%	99.95%	97.03%
	transPhyAtt	1.61%	5.00%	31.46%	58.44%	77.60%	34.82%
	MPA (Ours)	**98.85%**	**99.27%**	**99.84%**	**99.95%**	**99.95%**	**99.57%**
YOLO-World	Normal	53.23%	63.59%	69.53%	75.83%	81.72%	68.78%
	FCA	82.55%	89.17%	90.31%	90.73%	91.56%	88.86%
	transPhyAtt	44.27%	56.30%	62.19%	67.81%	71.98%	60.51%
	MPA (Ours)	**99.58%**	**99.90%**	**100.00%**	**100.00%**	**100.00%**	**99.90%**

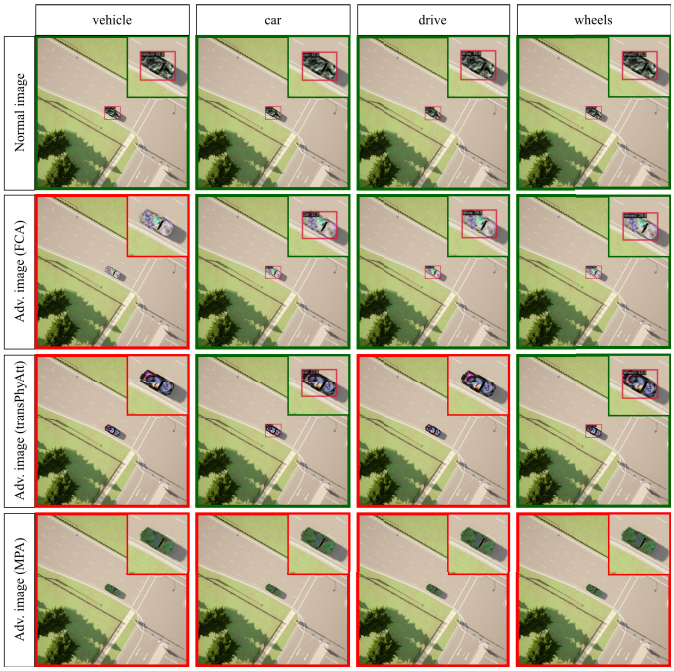


Fig. 3. Attack performance under different prompts using the GroundingDINO detector. Adversarial textures are trained, applied in simulation scenes, and evaluated with a confidence threshold of 0.1 and an IoU threshold of 0.3.

and exhibits a more natural visual appearance compared to FCA and transPhyAtt.

C. Viewpoint-Dependent Prompt Response Analysis

To quantify prompt response under viewpoint changes, we measure the prompt-conditioned detection confidence of GroundingDINO for a normally textured armored vehicle across 36 viewpoints. The viewpoints cover 12 azimuth angles and three elevation angles. At each viewpoint, the detector is queried separately with *vehicle*, *car*, *drive*, and *wheels*. For a given prompt, the response is defined as the highest confidence among predicted boxes whose IoU with the ground-truth box exceeds 0.3. No confidence threshold is applied.

Fig. 4 presents the heatmap of confidence scores for every prompt and viewpoint combination. The responses to *vehicle* and *car* vary substantially across viewpoints, with standard

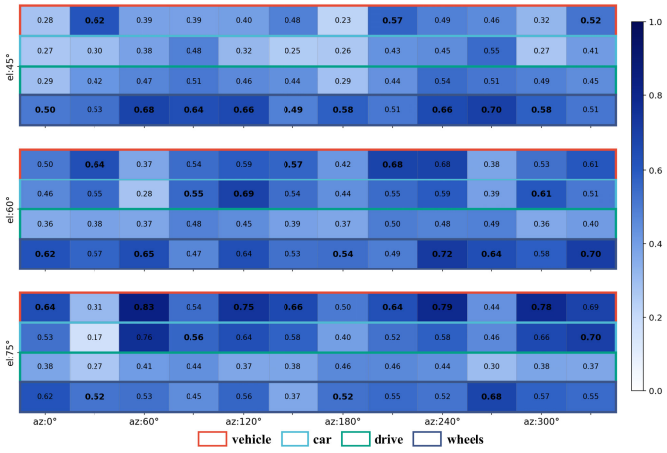


Fig. 4. Prompt-conditioned detection confidence of GroundingDINO for the same normally textured armored vehicle across 36 viewpoints spanning 12 azimuth angles and three elevation angles. Each cell reports the highest confidence among predicted boxes with IoU > 0.3 relative to the ground-truth box. Bold values mark the prompt with the highest confidence at each viewpoint.

deviations of 0.1494 and 0.1464. The responses to *drive* and *wheels* are more stable, with standard deviations of 0.0675 and 0.0807, but their relative strengths still change with viewing geometry. The prompt producing the highest confidence is viewpoint dependent: *vehicle* dominates in 15 of the 36 viewpoints, *wheels* in 16, *car* in 5, and *drive* in none.

These results show that prompt dominance is not fixed across viewpoints. As the viewing geometry changes, different structural and functional cues become visible and contribute differently to the detector response under each textual query. Consequently, optimizing against only one prompt can leave residual prompt-conditioned responses at viewpoints where another prompt provides stronger target evidence. Joint optimization over multiple prompts constrains these complementary responses together and provides a stronger basis for viewpoint-robust physical camouflage.

D. Multiview Robustness Analysis

To explicitly evaluate robustness under diverse observation conditions, we conduct a multiview robustness analysis by measuring attack performance across different viewing angles.

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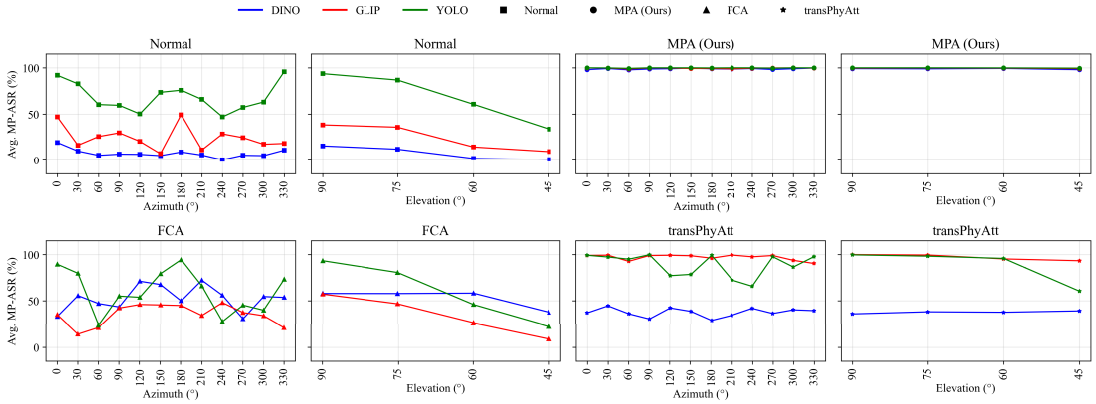


Fig. 5. Multiview attack performance across different viewing angles. For each viewpoint, the mean MP-ASR over five confidence thresholds is reported, illustrating robustness to viewpoint variation.

As shown in Fig. 5, the ASR is reported from multiple viewpoints, where the mean MP-ASR over five confidence thresholds is adopted as the evaluation metric, consistent with the protocol used in Table II.

From the quantitative results in Fig. 5, prior methods exhibit noticeable performance fluctuations across viewing angles, indicating a strong sensitivity to changes in observation perspective. In contrast, the proposed method maintains consistently higher MP-ASR across all evaluated viewpoints, demonstrating improved robustness to viewpoint variation. This behavior reflects a more stable adversarial effect under continuously changing observation conditions.

Such robustness can be attributed to the combined effect of dense multiview data acquisition and multiprompt joint optimization. The dense viewpoint sampling exposes the optimization process to diverse perspectives, while the multiprompt loss mitigates prompt-sensitive semantic instability, jointly stabilizing adversarial effectiveness across viewpoints.

In addition to the quantitative evaluation, Fig. 6 presents representative qualitative results across a range of viewing angles. As illustrated, the proposed method consistently succeeds across the majority of angle–prompt combinations, providing complementary visual evidence of its robustness under viewpoint variation.

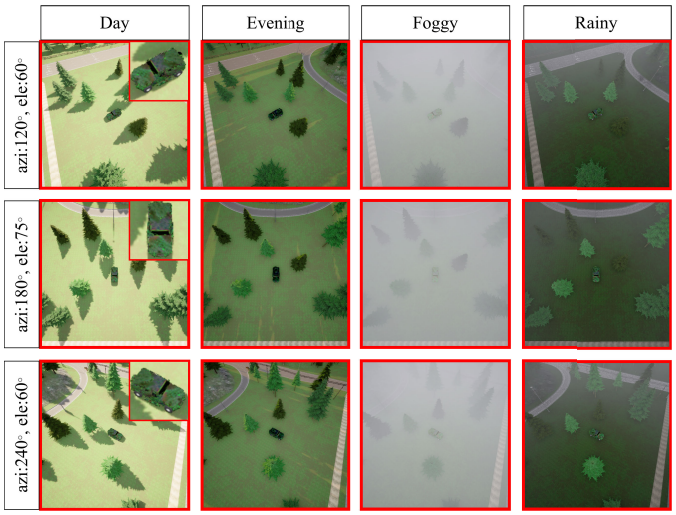


Fig. 6. Qualitative visualization of adversarial attack effectiveness under diverse weather conditions and viewing angles. The columns correspond to different time-of-day and weather conditions, including daytime, evening, foggy, and rainy scenes, while the rows represent different camera viewpoints with varying azimuth and elevation angles. Despite significant appearance changes induced by weather and viewpoint variations, the proposed adversarial camouflage consistently suppresses target detection across all settings.

E. Transferability and Robustness Across Diverse Scenes and Target Types

To systematically assess the generalization capability of the proposed method, this section investigates its transferability and robustness under variations in scene backgrounds and vehicle target categories. All experiments are conducted under a unified framework, where the overall optimization pipeline and training procedure are kept unchanged, enabling a controlled analysis of the effects induced solely by scene and target variations.

1) Experimental Settings: To evaluate robustness across diverse conditions, we construct multiple target–scene configurations within a unified experimental framework. Except for changes in scene context and vehicle category, all experimental configurations strictly follow those described in Sections III and IV-A, including the model architecture, optimization strategy, loss functions, and training and evaluation protocols. This design ensures fair and comparable results across different experimental settings.

Regarding dataset construction, in addition to the original armored vehicle remote sensing dataset collected in forest environments, we further build a civilian vehicle dataset captured under urban scenes. The civilian vehicle dataset follows the same data acquisition pipeline, viewpoint sampling configuration, and image resolution settings as the armored vehicle dataset. Specifically, 200 vehicle trajectory points are sampled in urban environments, with each trajectory point providing 48 multiview images. During preprocessing, samples with severe occlusion or insufficient visibility are filtered out. The resulting dataset contains 9503 remote sensing images with a resolution of 800 × 800 pixels, among which 7602

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TABLE III

QUANTITATIVE EVALUATION OF ADVERSARIAL PERFORMANCE FOR ARMORED VEHICLES WITH DESERT-STYLE CAMOUFLAGE. THIS EXPERIMENT ASSESSES THE ROBUSTNESS OF THE PROPOSED METHOD UNDER NONGREEN CAMOUFLAGE INSTANTIATIONS. ALL METHODS FOLLOW THE SAME OPTIMIZATION AND EVALUATION PROTOCOL AS THE BASELINE SETTING; THE DESERT-STYLE COLOR CONSTRAINT IS APPLIED ONLY TO MPA, WHEREAS THE BASELINE METHODS ARE OPTIMIZED WITH THE DEFAULT COLOR CONSTRAINT

Model	Method	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
GroundingDINO	Normal	0.00%	0.00%	0.36%	6.46%	30.94%	7.55%
	FCA	0.00%	0.26%	16.67%	73.54%	97.92%	37.68%
	transPhyAtt	0.36%	26.20%	59.11%	84.64%	95.26%	53.11%
	MPA (Ours)	**78.23%**	**87.29%**	**92.50%**	**92.66%**	**96.30%**	**89.40%**
GLIP	Normal	3.54%	4.38%	15.73%	35.73%	62.81%	24.44%
	FCA	90.57%	94.84%	99.84%	99.95%	99.95%	97.03%
	transPhyAtt	1.61%	5.00%	31.46%	58.44%	77.60%	34.82%
	MPA (Ours)	**95.31%**	**95.78%**	**96.15%**	**96.35%**	**96.46%**	**96.01%**
YOLO-World	Normal	53.23%	63.59%	69.53%	75.83%	81.72%	68.78%
	FCA	82.55%	89.17%	90.31%	90.73%	91.56%	88.86%
	transPhyAtt	44.27%	56.30%	62.19%	67.81%	71.98%	60.51%
	MPA (Ours)	**93.91%**	**98.02%**	**98.54%**	**98.65%**	**98.80%**	**97.58%**

images are used for training and 1901 images are reserved for testing.

In terms of scene variation, the urban dataset represents a nongreen background shift relative to the forest environment used in the baseline experiments. To further examine whether the proposed method is tied to a specific camouflage color, we additionally instantiate the color constraint using a desert-style color range by adjusting the parameters $c_{\min}$ and $c_{\max}$. Importantly, this adjustment only affects the allowable color distribution of the generated textures, while the optimization objectives and learning process remain unchanged. This design allows us to disentangle the effects of background variation and color prior selection without introducing additional methodological changes.

Based on the above configurations, we evaluate three additional settings beyond the baseline armored-vehicle–forest scenario: armored vehicles with desert-style camouflage, civilian vehicles with desert-style camouflage, and civilian vehicles with forest-style camouflage. These settings enable a systematic evaluation of robustness to target type variation, nongreen camouflage instantiations, and the flexibility of the color prior under a unified optimization framework.

2) Results and Analysis: This section reports the adversarial performance under the newly introduced configurations, including robustness to different target types, nongreen urban backgrounds, and desert-style color constraints. Fig. 7 provides qualitative visualizations of the learned adversarial camouflage under these previously unseen settings, illustrating consistent attack effectiveness across different vehicle categories, scene contexts, and camouflage color instantiations. The quantitative results are summarized in Tables III–V.

We first analyze robustness under nongreen background conditions by evaluating civilian vehicles in urban scenes. Compared with the baseline forest setting, the proposed method maintains strong adversarial effectiveness under urban backgrounds, indicating that MPA does not rely on green vegetation statistics and generalizes to visually distinct scene distributions.

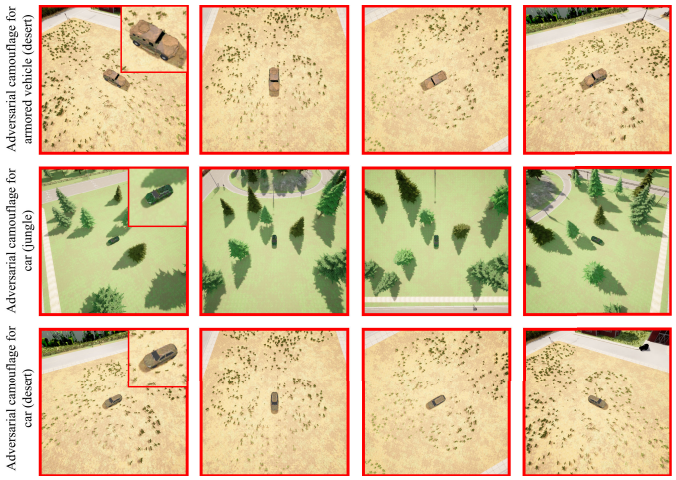


Fig. 7. Qualitative visualizations of the learned adversarial camouflage under previously unseen configurations, including civilian vehicles in urban scenes and adversarial camouflage instantiated with desert-style color constraints.

We further examine robustness to different target types by comparing adversarial performance between armored and civilian vehicles, where the camouflage texture is independently optimized for each vehicle category. Despite notable differences in geometric structure and visual appearance between these vehicle categories, the proposed method preserves satisfactory adversarial effectiveness, suggesting that the multiprompt-driven optimization captures shared semantic vulnerabilities across vehicle concepts.

Finally, we analyze the impact of camouflage color instantiation by applying desert-style color constraints to both armored and civilian vehicles. The results show that changing the camouflage tone does not significantly affect attack effectiveness, demonstrating that the proposed framework is not tied to a specific color prior and supports flexible, user-configurable camouflage instantiation.

Taken together, these results demonstrate that the proposed approach exhibits robust adaptability across nongreen urban backgrounds, diverse vehicle target types, and different camouflage color instantiations, supporting its applicability in realistic remote sensing scenarios.

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TABLE IV

QUANTITATIVE EVALUATION OF ADVERSARIAL PERFORMANCE FOR CIVILIAN VEHICLES WITH FOREST-STYLE CAMOUFLAGE. THE ADVERSARIAL TEXTURE IS INDEPENDENTLY OPTIMIZED ON THE CIVILIAN VEHICLE UNDER THE SAME TRAINING AND EVALUATION PROTOCOL AS THE DEFAULT EXPERIMENT

Model	Method	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
GroundingDINO	Normal	2.63%	3.58%	7.94%	22.41%	41.40%	15.59%
	FCA	2.52%	21.30%	74.91%	96.42%	99.63%	58.96%
	transPhyAtt	2.58%	8.10%	29.14%	58.39%	82.06%	36.05%
	MPA (Ours)	**94.32%**	**99.74%**	**100.00%**	**100.00%**	**100.00%**	**98.81%**
GLIP	Normal	5.79%	7.05%	14.31%	22.09%	36.61%	17.17%
	FCA	79.33%	92.16%	99.11%	99.95%	100.00%	94.11%
	transPhyAtt	7.36%	10.99%	24.78%	45.92%	65.33%	30.88%
	MPA (Ours)	**95.79%**	**97.95%**	**99.74%**	**100.00%**	**100.00%**	**98.70%**
YOLO-World	Normal	40.08%	52.18%	60.60%	67.23%	72.75%	58.57%
	FCA	91.69%	97.90%	99.26%	99.63%	99.79%	97.65%
	transPhyAtt	64.54%	77.59%	83.80%	87.01%	89.22%	80.43%
	MPA (Ours)	**99.63%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**99.93%**

TABLE V

QUANTITATIVE EVALUATION OF ADVERSARIAL PERFORMANCE FOR CIVILIAN VEHICLES WITH DESERT-STYLE CAMOUFLAGE. THE ADVERSARIAL TEXTURE OF MPA IS INDEPENDENTLY OPTIMIZED ON THE CIVILIAN VEHICLE WITH A DESERT-STYLE COLOR CONSTRAINT, WHEREAS THE BASELINE METHODS ARE OPTIMIZED WITH THE DEFAULT COLOR CONSTRAINT, COMBINING TARGET TYPE VARIATION AND NONGREEN CAMOUFLAGE INSTANTIATION UNDER THE SAME OPTIMIZATION FRAMEWORK AND LOSS DESIGN

Model	Method	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
GroundingDINO	Normal	2.63%	3.58%	7.94%	22.41%	41.40%	15.59%
	FCA	2.52%	21.30%	74.91%	96.42%	99.63%	58.96%
	transPhyAtt	2.58%	8.10%	29.14%	58.39%	82.06%	36.05%
	MPA (Ours)	**75.22%**	**98.00%**	**100.00%**	**100.00%**	**100.00%**	**94.64%**
GLIP	Normal	5.79%	7.05%	14.31%	22.09%	36.61%	17.17%
	FCA	79.33%	92.16%	99.11%	99.95%	100.00%	94.11%
	transPhyAtt	7.36%	10.99%	24.78%	45.92%	65.33%	30.88%
	MPA (Ours)	**96.74%**	**98.42%**	**99.79%**	**99.95%**	**100.00%**	**98.98%**
YOLO-World	Normal	40.08%	52.18%	60.60%	67.23%	72.75%	58.57%
	FCA	91.69%	97.90%	99.26%	99.63%	99.79%	97.65%
	transPhyAtt	64.54%	77.59%	83.80%	87.01%	89.22%	80.43%
	MPA (Ours)	**99.26%**	**99.79%**	**100.00%**	**100.00%**	**100.00%**	**99.81%**

F. Physical Adversarial Attack

In this section, we extend our evaluation to real-world scenarios. The learned adversarial textures are exported, printed, and manually affixed to 3-D-printed vehicle models for physical testing. Fig. 8 illustrates the detection results under different camouflage methods, where the adversarial textures are optimized against and evaluated on the GroundingDINO detector with a confidence threshold of 0.3. As shown, our approach produces adversarial patterns with lower visual saliency and achieves higher attack success across multiple textual prompts, demonstrating improved stealth and robustness in physical settings.

To further evaluate robustness under realistic physical environmental perturbations, we conduct additional physical experiments under diverse illumination and weather conditions. Specifically, adversarially camouflaged 3-D-printed vehicle models are captured under four representative real-world settings: midday, dusk, clear weather, and rainy weather. For each detector, we collect 24 images from different viewpoints under each condition, yielding 96 physical images per detector and 288 images across the three detectors. Following the same evaluation protocol, we compute the corresponding MP-ASR for each condition and report the overall Avg. MP-ASR as the final metric. For each detector, a separate adversarial texture is independently optimized against and evaluated on that same detector under the white-box setting.

The quantitative results are summarized in Table VI. The results show that the proposed MPA consistently maintains strong adversarial effectiveness across all evaluated physical conditions, demonstrating stable performance under variations in illumination and weather. These findings provide direct empirical evidence that the learned adversarial camouflage remains effective beyond controlled environments, supporting the practical feasibility of the proposed physical attack framework in real-world scenarios.

G. Adversarial Transfer Attack

In this section, we evaluate the transferability of MPA in three parts. First, we train and evaluate the attack with expanded prompt sets of eight and 12 prompt words to assess the method's robustness to an increased number of prompts. Second, we use full sentences as textual queries to evaluate the attack's effectiveness under different prompt formulations. Third, we perform cross-model transfer experiments to verify the proposed method's black-box attack capability.

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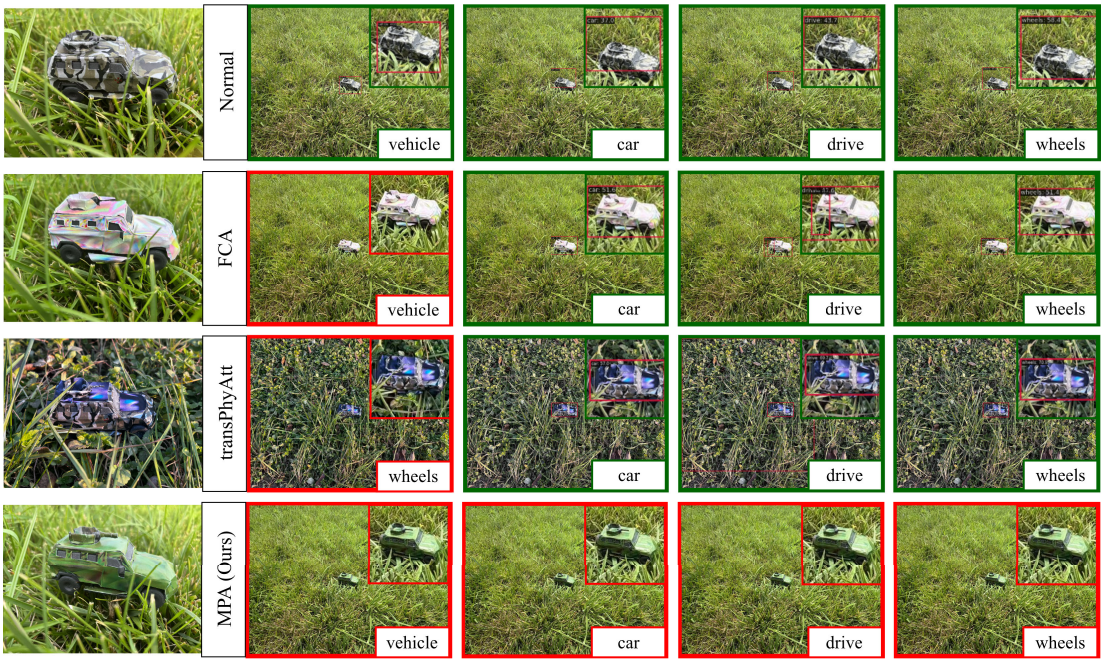


Fig. 8. Physical appearance and performance of adversarial camouflage applied to 3-D-printed vehicle models. The textures are generated by different methods and evaluated using the GroundingDINO detector (confidence threshold = 0.3).

1) Evaluation Under Expanded Prompt Sets: To evaluate the robustness of MPA to variations in textual prompts, we investigate its performance under expanded prompt sets of different sizes. Unless otherwise stated, the results reported in the main experiments are obtained using a four-prompt configuration. In this section, we explicitly vary the prompt set size and analyze its impact as an independent experimental factor.

We compute the MP-ASR across multiple confidence thresholds (0.1–0.5) and report the average. Table VII presents results obtained using an eight-prompt set (*vehicle, car, drive, wheels, transport, auto, automobile, and motor*), while Table VIII further extends this evaluation to a twelve-prompt set (*vehicle, car, drive, wheels, transport, auto, automobile, motor, truck, van, bus, and sedan*). These settings introduce increasingly stronger joint semantic constraints and therefore represent more challenging optimization conditions than the default four-prompt setting.

It is important to note that the results reported in Tables VII and VIII are not directly comparable to those in Table II, as they correspond to different prompt set sizes. As the number of prompts increases, the optimization difficulty of multiprompt adversarial training also increases, which can naturally lead to variations in MP-ASR.

Despite this increased difficulty, MPA consistently outperforms baseline methods at all confidence thresholds and maintains high ASRs across both expanded prompt sets. These results demonstrate that the learned adversarial camouflage generalizes well across different prompt vocabularies and remains effective even under more stringent multiprompt constraints.

2) Evaluation With Sentence-Level Prompts: Here, we evaluate the robustness of MPA under sentence-level prompts to test its generalization across different prompt formulations. Three representative cases are considered: *a vehicle covered in green camouflage*, *an army-green armored vehicle in the clearing*, and *a green military armored vehicle driving on the road*. The corresponding results, reported in Table IX using MP-ASR@0.5, show that MPA achieves consistently high ASRs even under more specific sentence-level queries, demonstrating effective transferability from word-level to sentence-level prompts.

3) Cross-Model Transfer Attack: To evaluate the transferability of MPA across different detectors, we conduct black-box experiments in which camouflage textures optimized on a source model are directly applied to other target

TABLE VI

PHYSICAL ATTACK PERFORMANCE UNDER DIVERSE REAL-WORLD ENVIRONMENTAL CONDITIONS. ADVERSARIALLY CAMOUFLAGED 3-D-PRINTED VEHICLE MODELS ARE EVALUATED UNDER FOUR REPRESENTATIVE PHYSICAL SETTINGS: MIDDAY, DUSK, CLEAR WEATHER, AND RAINY WEATHER. FOR EACH DETECTOR, 24 IMAGES CAPTURED FROM DIFFERENT VIEWPOINTS UNDER EACH CONDITION ARE USED TO COMPUTE THE CORRESPONDING MP-ASR, YIELDING 96 IMAGES PER DETECTOR AND 288 IMAGES ACROSS THE THREE DETECTORS. THE OVERALL AVG. MP-ASR IS REPORTED AS THE FINAL METRIC. A SEPARATE ADVERSARIAL TEXTURE IS INDEPENDENTLY OPTIMIZED AGAINST AND EVALUATED ON EACH DETECTOR UNDER THE WHITE-BOX SETTING

Model \ Method	Normal	FCA	transPhyAtt	MPA (Ours)
GroundingDINO	14.84%	42.64%	31.41%	**98.56%**
GLIP	15.83%	57.61%	72.91%	**99.45%**
YOLO-World	41.98%	75.06%	79.34%	**99.63%**

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TABLE VII

MP-ASR EVALUATION ON THREE OPEN-VOCABULARY DETECTORS UNDER AN EXPANDED EIGHT-PROMPT SETTING. THIS EXPERIMENT EVALUATES ROBUSTNESS TO ENLARGED PROMPT VOCABULARIES AND IS NOT DIRECTLY COMPARABLE TO THE DEFAULT FOUR-PROMPT SETTING REPORTED IN TABLE II

Model	Method	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
GroundingDINO	Normal	0.00%	0.00%	0.00%	0.05%	2.29%	0.47%
	FCA	0.00%	0.00%	0.94%	16.82%	67.92%	17.14%
	transPhyAtt	0.00%	1.20%	10.10%	29.48%	54.17%	18.99%
	MPA (Ours)	**85.31%**	**98.96%**	**99.84%**	**99.84%**	**100.00%**	**96.79%**
GLIP	Normal	2.50%	3.23%	14.64%	32.08%	52.76%	21.04%
	FCA	90.16%	94.38%	99.84%	99.95%	99.95%	96.86%
	transPhyAtt	4.48%	11.51%	49.01%	77.81%	92.50%	47.06%
	MPA (Ours)	**97.03%**	**98.49%**	**99.90%**	**99.95%**	**99.95%**	**99.06%**
YOLO-World	Normal	53.18%	63.54%	68.85%	74.43%	79.01%	67.80%
	FCA	79.32%	87.86%	90.16%	90.73%	91.56%	87.93%
	transPhyAtt	47.50%	61.51%	67.66%	72.08%	76.82%	65.11%
	MPA (Ours)	**97.08%**	**99.48%**	**99.90%**	**99.95%**	**99.95%**	**99.27%**

TABLE VIII

MP-ASR EVALUATION ON THREE OPEN-VOCABULARY DETECTORS UNDER A MORE CHALLENGING TWELVE-PROMPT SETTING. INCREASING THE NUMBER OF PROMPTS INTRODUCES STRONGER JOINT SEMANTIC CONSTRAINTS AND HIGHER OPTIMIZATION DIFFICULTY COMPARED TO THE DEFAULT FOUR-PROMPT CONFIGURATION IN TABLE II

Model	Method	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
GroundingDINO	Normal	0.00%	0.00%	0.00%	0.05%	2.24%	0.46%
	FCA	0.00%	0.00%	0.83%	16.35%	67.40%	16.92%
	transPhyAtt	0.00%	0.57%	5.52%	16.35%	29.95%	10.48%
	MPA (Ours)	**77.55%**	**97.97%**	**99.64%**	**99.84%**	**99.95%**	**94.99%**
GLIP	Normal	2.45%	3.18%	14.64%	32.03%	52.76%	21.01%
	FCA	90.16%	94.38%	99.84%	99.95%	99.95%	96.86%
	transPhyAtt	4.48%	11.56%	49.01%	77.81%	92.50%	47.07%
	MPA (Ours)	**97.03%**	**98.49%**	**99.90%**	**99.95%**	**99.95%**	**99.06%**
YOLO-World	Normal	27.03%	37.34%	46.77%	56.46%	65.36%	46.59%
	FCA	42.66%	56.77%	64.27%	69.27%	74.48%	61.49%
	transPhyAtt	24.58%	35.21%	42.6%	49.17%	57.76%	41.86%
	MPA (Ours)	**88.91%**	**96.04%**	**97.86%**	**98.75%**	**99.32%**	**96.18%**

TABLE IX

PROMPT-TO-SENTENCE TRANSFERABILITY: MP-ASR@0.5 FOR THREE SENTENCE-LEVEL PROMPTS. S1: A VEHICLE COVERED IN GREEN CAMOUFLAGE; S2: AN ARMY-GREEN ARMORED VEHICLE IN THE CLEARING; S3: A GREEN MILITARY ARMORED VEHICLE DRIVING ON THE ROAD

Prompt \ Model	GroundingDINO	GLIP	YOLO-World
S1	100.00%	99.80%	99.80%
S2	100.00%	100.00%	100.00%
S3	91.64%	91.00%	91.10%

models. The results are reported in Table X, where diagonal entries correspond to white-box attacks (i.e., textures evaluated on their source model), and off-diagonal entries represent black-box transfer attacks. All results are measured using MP-ASR@0.5. As shown, although the textures are learned in a white-box setting, they retain strong attack effectiveness when transferred to other models, indicating robust cross-model transferability. Among the models, textures trained on GroundingDINO exhibit the highest transferability, while those trained on YOLO-World transfer least effectively, with GLIP showing intermediate performance. This phenomenon may be attributed to differences in model architecture and training paradigms: GroundingDINO employs stronger cross-modal alignment and richer pretraining on grounding data, which facilitates generalization across detectors, whereas YOLO-World relies on the YOLO architecture with comparatively limited vision–language coupling, making its learned camouflage patterns less transferable.

H. Saliency Study

To evaluate the visual perceptibility of the adversarial camouflage generated by the proposed method, we adopt two widely used saliency detection algorithms: spectral residual saliency detection [54] and fine-grained saliency detection [55]. In the experiments, we select two camera heights (2 and 4 m) and capture rendered images from eight different viewpoints at each height. Both saliency detection methods are applied to all rendered images to assess the visual conspicuity of the camouflage under varying viewing angles and camera heights. Finally, the average saliency values are computed and used as comparative metrics.

As shown in Table XI, our method consistently yields lower visual saliency across all models, highlighting its effectiveness in suppressing perceptual cues and enhancing practical applicability.

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TABLE X

CROSS-MODEL TRANSFER RESULTS. DIAGONAL ENTRIES DENOTE WHITE-BOX ATTACKS; OFF-DIAGONAL ENTRIES DENOTE BLACK-BOX TRANSFER ATTACKS. VALUES ARE MP-ASR@0.5

Camouflage \ Model	GroundingDINO	GLIP	YOLO-World
GroundingDINO	100.00%	98.07%	99.74%
GLIP	10.68%	99.95%	95.89%
YOLO-World	15.57%	54.43%	100.00%

TABLE XI

QUANTITATIVE VISUAL SALIENCY RESULTS ACROSS DIFFERENT METHODS. HERE, A REPRESENTS THE SPECTRAL RESIDUAL SALIENCY DETECTION ALGORITHM, WHILE B REFERS TO THE FINE-GRAINED SALIENCY DETECTION ALGORITHM. THE REPORTED VALUES ARE THE MEAN SALIENCY SCORES COMPUTED ACROSS 8 VIEWPOINTS AT TWO CAMERA HEIGHTS

Model	Method	A↓	B↓
GroundingDINO	FCA	25.07	61.47
	transPhyAtt	23.61	58.36
	MPA (Ours)	**19.97**	**51.07**
GLIP	FCA	23.53	58.54
	transPhyAtt	25.97	64.11
	MPA (Ours)	**22.02**	**58.37**
YOLO-World	FCA	22.54	65.66
	transPhyAtt	24.75	62.47
	MPA (Ours)	**22.04**	**59.47**

TABLE XII

MP-ASR UNDER DIFFERENT LOSS TERMS

Method	Avg. MP-ASR↑		
	GroundingDINO	GLIP	YOLO-World
$\mathcal{L}_{IoU}$	37.76%	38.14%	78.16%
$\mathcal{L}_{multi}$	71.93%	98.92%	99.52%
$\mathcal{L}_{IoU} + \mathcal{L}_{multi}$	**98.88%**	**99.57%**	**99.90%**

I. Ablation Study

In this part of the study, we analyze how different loss components and the color constraint function influence the effectiveness of the adversarial camouflage.

1) Effect of Different Loss Terms: We compare the attack performance under different loss configurations by testing across five confidence thresholds (0.1–0.5) and reporting the average as the evaluation metric. The impact of each loss term is shown in Table XII, where $\mathcal{L}_{IoU}$ denotes the IoU-based confidence loss, and $\mathcal{L}_{multi}$ represents the MPA loss. The best attack performance is achieved when both loss terms are applied simultaneously.

2) Prompt Inference Strategy: We further analyze whether the multiprompt loss should be computed using joint prompt inference or separate prompt-conditioned inference. Joint inference evaluates multiple prompts in one detector call and is computationally cheaper, whereas separate inference evaluates each prompt independently and sums the corresponding losses. For adversarial optimization, the central issue is whether joint inference preserves the same prompt-conditioned confidence scores as independent inference. If joint inference changes the confidence scale of individual prompts, then the resulting attack loss may no longer accurately represent the detector response induced by each textual query.

We use GroundingDINO for this controlled comparison because its public implementation provides a clear multiphrase inference interface, allowing the two strategies to be compared under the same detector, image, prompts, and detection parameters. This experiment analyzes the effect of inference strategy on confidence scores and optimization behavior. The final MPA implementation applies separate prompt-conditioned inference consistently to all three evaluated detectors, including GLIP, GroundingDINO, and YOLO-World.

Fig. 9 compares the two inference strategies on the same image. Under joint inference, the confidence scores for *car*, *drive*, *vehicle*, and *wheels* are 25.7%, 10.6%, 28.8%, and 14.7%, respectively. Independent inference of the same prompts yields 46.6%, 24.2%, 47.0%, and 34.3%. This comparison shows that joint prompt inference and separate prompt-conditioned inference do not produce confidence scores on the same scale. In this case, joint inference suppresses the confidence values of semantically related prompts, which reduces the loss magnitude available for texture optimization.

We further train two MPA variants under identical conditions to quantify the effect on adversarial optimization. The joint-inference variant passes all four phrases in one detector call and computes the loss from the joint outputs. The separate-inference variant evaluates the four prompts independently and sums their prompt-conditioned losses. Both variants use the same texture initialization, data split, renderer, prompt set, color constraint, loss formulation, optimizer, optimization budget, and evaluation protocol.

As shown in Table XIII, separate inference achieves an Avg. MP-ASR of 98.88%, compared with 93.38% for joint inference. The difference is largest at the strictest confidence threshold, where MP-ASR increases from 69.10% to 94.90%. These results indicate that joint inference can weaken the optimization signal when target responses have moderate confidence, while separate prompt-conditioned inference preserves stronger query-specific gradients for multiprompt camouflage optimization.

3) Sensitivity to the IoU Threshold: The training and evaluation IoU thresholds affect different stages of the proposed framework. The training threshold $\tau_{\text{IoU}}^{\text{train}}$ determines which predictions are treated as target-associated responses in the loss. The evaluation threshold $\tau_{\text{IoU}}^{\text{eval}}$ determines whether a prediction is counted as a valid detection when computing MP-ASR. To analyze the sensitivity to both choices, we conduct a joint threshold study on GroundingDINO with $\tau_{\text{IoU}}^{\text{train}}, \tau_{\text{IoU}}^{\text{eval}} \in \{0.1, 0.3, 0.5, 0.7\}$. For each training threshold, an MPA texture is independently optimized under the same data split, renderer, prompt set, optimization budget, and color constraint. Each texture is then evaluated under all four evaluation thresholds and five confidence thresholds from 0.1 to 0.5.

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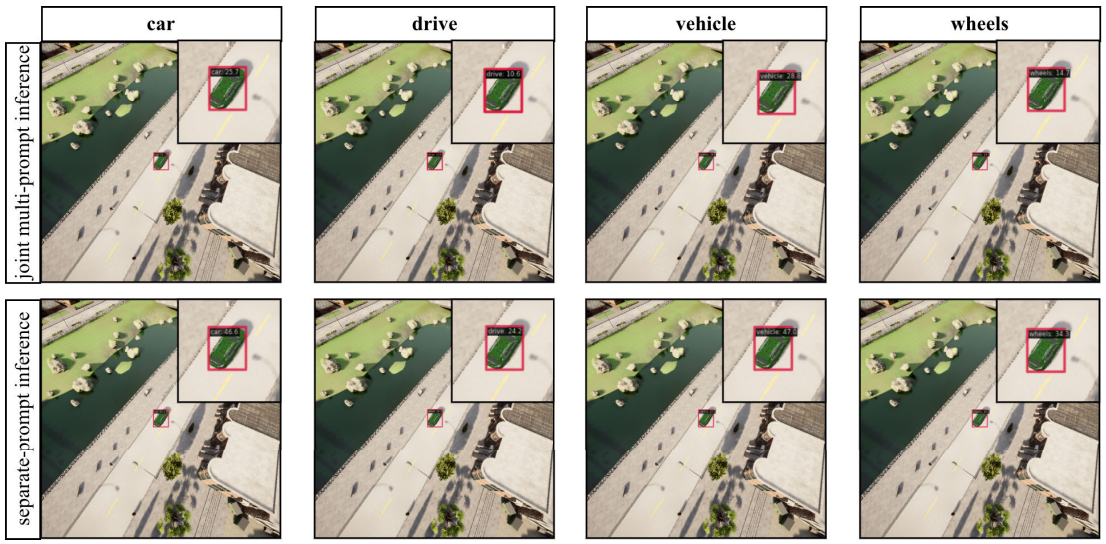


Fig. 9. Comparison between joint and separate prompt inference on GroundingDINO. The first row shows detections from one joint call with four prompts. The second row shows detections from independent evaluation of each prompt. All detection parameters are unchanged.

TABLE XIII

ABLATION OF PROMPT INFERENCE STRATEGIES ON GROUNDINGDINO. BOTH VARIANTS USE THE SAME OPTIMIZATION AND EVALUATION PROTOCOL

Strategy	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
Joint inference	69.10%	97.93%	99.89%	100.00%	100.00%	93.38%
Separate inference	94.90%	99.69%	99.84%	99.95%	100.00%	98.88%

TABLE XIV

SENSITIVITY OF MPA TO TRAINING AND EVALUATION IOU THRESHOLDS ON GROUNDINGDINO. VALUES ARE MP-ASR (%) UNDER DIFFERENT CONFIDENCE THRESHOLDS AND AVG. MP-ASR (%)

$\tau_{\text{IoU}}^{\text{train}}$	$\tau_{\text{IoU}}^{\text{eval}}$	Conf.=0.1	Conf.=0.2	Conf.=0.3	Conf.=0.4	Conf.=0.5	Avg.
0.1	0.1	86.89	99.56	100.00	100.00	100.00	97.29
0.1	0.3	89.74	99.62	100.00	100.00	100.00	97.87
0.1	0.5	90.49	99.67	100.00	100.00	100.00	98.03
0.1	0.7	92.10	99.67	100.00	100.00	100.00	98.35
0.3	0.1	76.63	99.29	100.00	100.00	100.00	95.18
0.3	0.3	94.17	99.73	100.00	100.00	100.00	98.78
0.3	0.5	94.86	99.73	100.00	100.00	100.00	98.92
0.3	0.7	95.94	99.73	100.00	100.00	100.00	99.13
0.5	0.1	75.34	98.96	100.00	100.00	100.00	94.86
0.5	0.3	88.38	99.56	100.00	100.00	100.00	97.59
0.5	0.5	89.46	99.62	100.00	100.00	100.00	97.82
0.5	0.7	91.56	99.62	100.00	100.00	100.00	98.24
0.7	0.1	65.73	97.22	99.94	100.00	100.00	92.58
0.7	0.3	85.47	98.09	99.94	100.00	100.00	96.70
0.7	0.5	87.03	98.20	99.94	100.00	100.00	97.03
0.7	0.7	92.75	99.51	100.00	100.00	100.00	98.45

Table XIV shows that MPA is stable across a wide range of evaluation IoU thresholds. With the default training threshold $\tau_{\text{IoU}}^{\text{train}} = 0.3$, Avg. MP-ASR remains above 95% for all evaluation thresholds and reaches 98.78% under the default evaluation threshold $\tau_{\text{IoU}}^{\text{eval}} = 0.3$. The same setting achieves 94.17% MP-ASR at the strictest confidence threshold of 0.1 and 100.00% MP-ASR when the confidence threshold is 0.3 or higher. For evaluation thresholds of 0.3 and higher, $\tau_{\text{IoU}}^{\text{train}} = 0.3$ gives the highest Avg. MP-ASR among the four training thresholds. Lower training thresholds include weakly overlapping predictions and dilute the optimization signal, whereas higher training thresholds remove moderately localized target predictions and reduce the available gradients. These results support the default use of 0.3 for target-associated prediction filtering.

4) Computational Efficiency: We further report the computational cost of MPA during offline texture optimization. This analysis is separated from deployment-time inference because MPA optimizes a surface texture before deployment. During optimization, each step includes differentiable rendering, prompt-conditioned detector inference, loss computation, backpropagation, and texture update. The victim detector remains frozen, and the adversarial texture is the only optimized variable.

Efficiency is measured on an NVIDIA RTX 4090 GPU with 24 GB memory. For each detector and vehicle configuration, the first ten optimization steps are used for GPU warm-up, followed by 100 measured steps. CUDA synchronization is applied before and after each timed step. The measured time therefore corresponds to the end-to-end cost of one optimization step. Peak GPU memory is recorded as the maximum allocated memory during optimization.

Table XV reports the runtime and memory consumption for two vehicle types and three open-vocabulary detectors. The civilian-vehicle texture contains 14 997 960 optimized parameters, whereas the armored-vehicle texture contains 49 285 584 optimized parameters. The armored vehicle therefore requires more memory and slightly longer optimization time across all detectors. YOLO-World is the most efficient victim detector in this optimization loop, requiring 0.1777 s/step for the civilian vehicle and 0.2093 s/step for the armored vehicle.

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TABLE XV

COMPUTATIONAL EFFICIENCY OF MPA DURING OFFLINE TEXTURE OPTIMIZATION. RUNTIME AND PEAK MEMORY ARE MEASURED ON ONE NVIDIA RTX 4090 GPU WITH 24 GB MEMORY

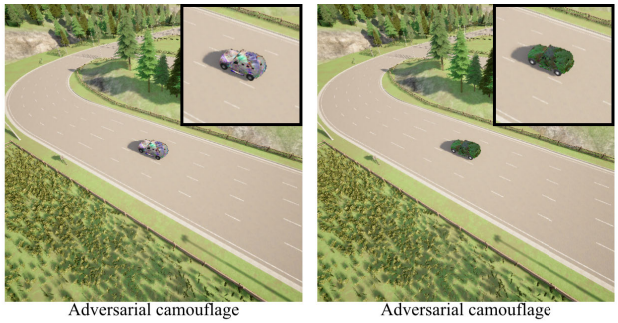
Vehicle Type	Detector	Optimized Parameters	Time/Step (s)	Peak Memory (GB)
Civilian vehicle	GroundingDINO	14,997,960	0.7728	14.62
	GLIP	14,997,960	0.7834	9.49
	YOLO-World	14,997,960	0.1777	8.27
Armored vehicle	GroundingDINO	49,285,584	0.8345	15.36
	GLIP	49,285,584	0.8954	12.40
	YOLO-World	49,285,584	0.2093	10.11

These measurements characterize the offline optimization cost of generating the adversarial texture. After optimization, the camouflage is stored as a surface texture and introduces no additional detector parameters or computational modules. Applying the optimized texture does not alter the runtime of an individual prompt-conditioned detector query. When several prompts are evaluated independently, the total evaluation cost scales with the number of prompts, as expected for multiquery open-vocabulary evaluation.

5) Effect of Color Constraint: In this section, we analyze the effect of the proposed color constraint by comparing adversarial camouflage generated with and without this constraint from both attack effectiveness and visual saliency perspectives. To evaluate attack performance, we measure the ASR under five confidence thresholds ranging from 0.1 to 0.5 and report the average value as the final metric, following the same evaluation protocol as in previous experiments. As reported in Table XVI, the adversarial camouflage generated with the color constraint achieves comparable and in some cases slightly higher MP-ASR than its unconstrained counterpart, indicating that the color constraint does not compromise attack effectiveness and may provide marginal performance gains.

Beyond ASR, we further assess the visual conspicuity of the generated camouflage using two complementary saliency detection methods, namely, spectral residual saliency detection and fine-grained saliency detection. We render images from eight different viewpoints and apply both saliency detection methods to evaluate how visually prominent the camouflage appears under varying observation conditions. The quantitative results in Table XVI show that the color-constrained camouflage consistently yields lower saliency scores across both detection methods, indicating reduced visual conspicuity and improved concealment.

Fig. 10 provides a qualitative comparison between adversarial camouflage generated with and without the color constraint under identical optimization settings. Although both variants are capable of suppressing detector responses, clear differences can be observed in their visual appearance. Without the color constraint, the optimized camouflage tends to exhibit irregular and high-contrast color patterns that deviate from natural scene statistics. In contrast, the color-constrained camouflage demonstrates more coherent color distributions and smoother transitions that better align with natural environmental appearance. This behavior is consistently observed across different viewpoints, leading to reduced saliency responses in both spectral residual and fine-grained saliency detection. These qualitative observations are consistent with the quantitative saliency metrics reported in Table XVI.



Adversarial camouflage ***without*** color constraint | Adversarial camouflage ***with*** color constraint

Fig. 10. Qualitative comparison of adversarial camouflage generated with and without the color constraint under identical optimization settings. While both variants achieve comparable MP-ASR, the color-constrained camouflage exhibits improved visual naturalness and reduced perceptual conspicuity, highlighting the role of the color constraint in enhancing physical plausibility rather than directly boosting ASR.

TABLE XVI

MP-ASR AND VISUAL SALIENCY SCORES UNDER DIFFERENT SETTINGS WITH AND WITHOUT THE COLOR CONSTRAINT. A REPRESENTS THE SPECTRAL RESIDUAL SALIENCY DETECTION ALGORITHM, WHILE B REFERS TO THE FINE-GRAINED SALIENCY DETECTION ALGORITHM

Model	Clamp	Avg. MP-ASR↑	A↓	B↓
GroundingDINO	w/o	97.73%	24.05	63.22
	w/	**98.88%**	**19.97**	**51.07**
GLIP	w/o	99.56%	23.57	65.12
	w/	**99.57%**	**22.02**	**58.37**
YOLO-World	w/o	99.83%	24.97	66.00
	w/	**99.90%**	**22.04**	**59.47**

Overall, these results indicate that the primary benefit of the color constraint lies in enhancing visual concealment and physical plausibility rather than substantially increasing the upper bound of ASR. From this perspective, the color constraint serves as an effective regularization mechanism that improves the deployability of adversarial camouflage by balancing attack effectiveness with perceptual inconspicuity, which is a critical requirement for physically realizable attacks in remote sensing scenarios.

6) Sensitivity to Color Interval Selection: We further analyze whether attack performance is sensitive to the selected color interval. The study is conducted on GroundingDINO under the same optimization and evaluation protocol. Four color settings are compared. The default setting uses the green interval adopted in the main experiments. The narrow setting reduces the width of this interval by half while preserving its center. The wide setting expands the interval width by 1.5 times, with values outside the valid RGB range clipped. The background-adaptive setting estimates the interval from local background statistics in the training scenes: RGB pixels are collected from annular regions around the target vehicle, and the 10th and 90th percentiles of each channel are used as the lower and upper bounds.

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TABLE XVII

SENSITIVITY ANALYSIS OF COLOR INTERVAL SELECTION ON GROUNDINGDINO. ALL SETTINGS USE THE SAME OPTIMIZATION AND EVALUATION PROTOCOL

Color Setting	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
Default	94.90%	99.69%	99.84%	99.95%	100.00%	98.88%
Narrow	72.50%	96.78%	99.71%	99.86%	100.00%	93.77%
Wide	95.84%	99.89%	100.00%	100.00%	100.00%	99.15%
Background-adaptive	81.36%	98.20%	99.83%	100.00%	100.00%	95.88%

TABLE XVIII

QUANTITATIVE COMPARISON OF DIFFERENT PROMPT STRATEGIES IN TERMS OF MP-ASR. THE RESULTS COMPARE CONFIDENCE-BASED SINGLE-PROMPT AUGMENTATION AND MULTIPROMPT OPTIMIZATION UNDER IDENTICAL EXPERIMENTAL SETTINGS. MULTIPROMPT OPTIMIZATION CONSISTENTLY YIELDS MORE THAN 5% AVERAGE MP-ASR IMPROVEMENT ACROSS ALL EVALUATED DETECTORS. ALL MODELS ARE TRAINED ON CIVILIAN VEHICLES UNDER FOREST SCENE CONDITIONS

Model	Strategy	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
GroundingDINO	Single-Prompt Augmentation	71.12%	88.79%	89.95%	90.37%	92.16%	86.48%
	Multi-Prompt	**94.32%**	**99.74%**	**100.00%**	**100.00%**	**100.00%**	**98.81%**
GLIP	Single-Prompt Augmentation	79.06%	82.53%	89.63%	89.95%	90.31%	86.30%
	Multi-Prompt	**95.79%**	**97.95%**	**99.74%**	**100.00%**	**100.00%**	**98.70%**
YOLO-World	Single-Prompt Augmentation	89.11%	90.00%	91.57%	92.72%	94.79%	91.64%
	Multi-Prompt	**99.63%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**99.93%**

Table XVII reports MP-ASR under five confidence thresholds and the average value. The default interval achieves 98.88% Avg. MP-ASR. The wide interval achieves the highest Avg. MP-ASR of 99.15%, which is expected because a broader color range gives the optimization more freedom. The narrow interval reduces Avg. MP-ASR to 93.77%, with the largest degradation at the strictest confidence threshold, indicating that overly restrictive color bounds can limit the adversarial texture. The background-adaptive interval achieves 95.88% Avg. MP-ASR and remains above 95% for confidence thresholds of 0.2 and higher. These results indicate that MPA remains effective under multiple color-range choices and that scene-statistics-based interval estimation is a feasible automatic alternative when a target camouflage palette is unavailable.

7) Effect of Prompt Strategies: This section investigates the impact of different prompt strategies on adversarial performance under identical experimental settings, with the goal of clarifying the necessity of the proposed multiprompt optimization design.

We compare three prompt strategies: single-prompt optimization, confidence-based single-prompt augmentation, and the proposed multiprompt optimization. For the prompt augmentation baseline, only one prompt is used in each forward pass, while the optimization prompt is dynamically selected from a predefined pool of semantically related synonyms. Specifically, for each iteration, detector responses under all candidate prompts are evaluated, and the prompt yielding the highest detection confidence is selected as the optimization direction. This strategy follows a prompt augmentation paradigm that adaptively prioritizes the most vulnerable semantic description at each iteration, without introducing joint semantic constraints across prompts.

Despite its adaptive nature, this prompt augmentation strategy is fundamentally different from multiprompt optimization. Prompt augmentation optimizes the adversarial camouflage with respect to a single prompt at a time, leading the optimization process to follow a sequence of locally dominant semantic directions. In contrast, multiprompt optimization enforces joint semantic constraints by aggregating losses across multiple prompts within the same optimization step. As a result, the adversarial camouflage is required to remain simultaneously effective under multiple semantic descriptions, which fundamentally alters the optimization geometry and reduces overfitting to any individual prompt.

The quantitative comparison between confidence-based single-prompt augmentation and multiprompt optimization is summarized in Table XVIII. Across all evaluated detectors, multiprompt optimization consistently achieves substantially higher MP-ASR than single-prompt augmentation, with improvements exceeding 5% in terms of average MP-ASR. This consistent performance gap demonstrates that the effectiveness of the proposed method cannot be attributed solely to increased prompt diversity or adaptive prompt selection, but rather stems from the joint semantic constraints imposed by multiprompt optimization.

It is worth noting that all results reported in Table XVIII are obtained from models trained on civilian vehicles under forest scene conditions, ensuring that the observed performance differences are solely induced by the prompt strategy rather than changes in data or training configuration.

8) Effect of Prompt Set Selection: This section examines the influence of prompt set selection on the adversarial performance of the proposed MPA framework under identical experimental settings, with the aim of evaluating its robustness to different prompt construction strategies.

In the main experiments, the prompt set consists of four semantically related terms (*vehicle, car, drive, and wheels*), which are intentionally chosen to capture complementary aspects of the target semantics, including category-level information, functional properties, and structural components. This manually constructed prompt set provides an intuitive and effective semantic description of the target and serves as

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TABLE XIX

ABLATION RESULTS ON THE EFFECT OF PROMPT SET SELECTION IN TERMS OF MP-ASR. THE RESULTS COMPARE A MANUALLY CONSTRUCTED PROMPT SET AND A WORDNET-BASED PROMPT SELECTION STRATEGY UNDER IDENTICAL EXPERIMENTAL SETTINGS. HIGH ADVERSARIAL PERFORMANCE IS CONSISTENTLY MAINTAINED ACROSS ALL EVALUATED DETECTORS. ALL MODELS ARE TRAINED ON THE URBAN-SCENE CIVILIAN VEHICLE DATASET

Model	Strategy	Prompt Set	MP-ASR@0.1↑	MP-ASR@0.2↑	MP-ASR@0.3↑	MP-ASR@0.4↑	MP-ASR@0.5↑	Avg. MP-ASR↑
GroundingDINO	Manual	vehicle, car, drive, wheels	94.32%	99.74%	100.00%	100.00%	100.00%	98.81%
	WordNet-based	car, auto, automobile, machine	96.37%	99.84%	100.00%	100.00%	100.00%	99.24%
GLIP	Manual	vehicle, car, drive, wheels	95.79%	97.95%	99.74%	100.00%	100.00%	98.70%
	WordNet-based	car, auto, automobile, machine	92.37%	94.53%	99.84%	100.00%	100.00%	97.35%
YOLO-World	Manual	vehicle, car, drive, wheels	99.63%	100.00%	100.00%	100.00%	100.00%	99.93%
	WordNet-based	car, auto, automobile, machine	99.74%	100.00%	100.00%	100.00%	100.00%	99.95%

a strong baseline throughout the article. To further assess whether the effectiveness of MPA depends on specific handcrafted prompt words, we introduce an alternative, systematic prompt set selection strategy based on WordNet.

Specifically, given a target concept, we construct a prompt set by retaining the target word itself and selecting three additional semantically related terms from WordNet according to semantic proximity. The selected terms include synonyms as well as closely related functional or attribute-level words. This procedure follows a clear and reproducible rule and does not rely on task-specific tuning or manual trial-and-error. Compared with manual prompt selection, the WordNet-based strategy offers a practical advantage by reducing reliance on expert knowledge while maintaining semantic coverage.

Under this setting, the MPA optimization pipeline, loss functions, and training configurations are kept unchanged, and only the prompt set is replaced according to the WordNet-based selection rule. All experiments are conducted using models trained on the urban-scene civilian vehicle dataset. The quantitative results are reported in Table XIX. As shown, MPA consistently maintains high MP-ASR when using the WordNet-based prompt set, with only marginal variations compared to the manually selected prompts across all evaluated detectors.

These results indicate that while the manually selected prompt set already yields strong adversarial performance, the effectiveness of MPA does not hinge on specific handcrafted prompt words. Instead, it arises from enforcing joint semantic constraints across multiple prompts. Consequently, the WordNet-based prompt selection strategy provides a convenient and effective alternative for deploying MPA in open-vocabulary scenarios, particularly when manual prompt design is less straightforward or unavailable.

V. ANALYSIS OF POTENTIAL DEFENSE COUNTERMEASURES

This section investigates potential defense countermeasures against the proposed MPA framework from the perspective of camouflaged object perception and responsible remote sensing security research. Rather than proposing new defense mechanisms, our objective is to assess the effectiveness of representative camouflage-aware detection models under adversarial camouflage conditions. Through this analysis, we aim to reveal current limitations of existing defenses and provide insights into promising directions for improving robustness in future remote sensing systems.

A. Defense Evaluation With COD Models

To analyze potential defense countermeasures against physical adversarial camouflage, we evaluate representative camouflaged object detection (COD) models under different camouflage conditions, including clean images, conventional camouflage textures, and adversarial camouflage generated by FCA, transPhyAtt, and the proposed MPA framework. This evaluation focuses on assessing the robustness of existing camouflage-aware detectors when confronted with physically realizable adversarial camouflage, rather than introducing or optimizing new defense strategies. The quantitative evaluation results are summarized in Table XX.

We select three representative COD methods for evaluation, namely, mutual graph learning (MGL), uncertainty-guided transformer reasoning (UGTR), and feature decomposition and edge reconstruction (FEDER). These models cover diverse architectural paradigms, including graph-based relational modeling, Transformer-based uncertainty reasoning, and feature decomposition with edge reconstruction, providing a representative view of current camouflage-aware perception techniques.

Following standard practice in COD, we report four commonly used evaluation metrics: mean absolute error (MAE, denoted as M), weighted F_β measure, mean enhanced-alignment measure (E_ϕ), and structure measure (S_α). From the perspective of COD performance, lower MAE and higher values of F_β, E_ϕ, and S_α indicate better detection quality. However, it is important to emphasize that the evaluation objective in this section differs fundamentally from that of conventional COD tasks. From the adversarial camouflage perspective, our goal is to suppress the effectiveness of camouflage-aware detectors. Accordingly, degraded COD performance—reflected by higher MAE and lower values of the remaining metrics in Table XX—corresponds to stronger camouflage effectiveness and more successful target concealment.

As shown in Table XX, a consistent performance degradation trend is observed across all evaluated COD models as the camouflage strength increases. Clean images yield the strongest COD performance, particularly for armored vehicles, which exhibit more stable appearance patterns and clearer structural cues. Conventional camouflage textures already

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TABLE XX

DEFENSE EVALUATION OF REPRESENTATIVE COD MODELS UNDER DIFFERENT CAMOUFLAGE CONDITIONS, INCLUDING CLEAN IMAGES, CONVENTIONAL CAMOUFLAGE, AND ADVERSARIAL CAMOUFLAGE GENERATED BY FCA, TRANSPHYATT, AND MPA. WE REPORT MAE (M), WEIGHTED F_β MEASURE, MEAN ENHANCED-ALIGNMENT MEASURE (E_ϕ), AND STRUCTURE MEASURE (S_α), WHERE THE ARROWS IN THE TABLE HEADER MARK THE DIRECTION OF STRONGER CAMOUFLAGE EFFECTIVENESS. LOWER MAE AND HIGHER VALUES OF THE OTHER METRICS INDICATE BETTER COD PERFORMANCE, WHILE WORSE COD PERFORMANCE CORRESPONDS TO STRONGER CAMOUFLAGE EFFECTIVENESS IN THE ATTACK SETTING

Target Category	Method	UGTR				MGL				FEDER			
		M↑	$F_\beta\downarrow$	$E_\phi\downarrow$	$S_\alpha\downarrow$	M↑	$F_\beta\downarrow$	$E_\phi\downarrow$	$S_\alpha\downarrow$	M↑	$F_\beta\downarrow$	$E_\phi\downarrow$	$S_\alpha\downarrow$
Car	Normal	0.012	0.043	0.553	0.533	0.013	0.046	0.532	0.528	0.005	0.416	0.744	0.747
	FCA	0.012	0.129	0.558	0.585	0.011	0.114	0.576	0.588	0.005	0.467	0.759	0.775
	transPhyAtt	0.013	0.061	0.531	0.538	0.012	0.048	0.554	0.537	0.004	0.509	0.784	0.798
	MPA (forest)	0.013	**0.008**	0.525	**0.499**	**0.013**	**0.004**	0.549	**0.495**	**0.009**	**0.197**	**0.657**	**0.620**
	MPA (desert)	**0.013**	0.022	**0.525**	0.507	0.012	0.009	**0.549**	0.501	0.008	0.324	0.684	0.700
Armored	Normal	0.004	0.638	0.834	0.844	0.006	0.450	0.735	0.773	0.002	0.762	0.931	0.900
	FCA	0.004	0.684	0.860	0.873	0.005	0.568	0.805	0.832	0.001	0.824	0.955	0.924
	transPhyAtt	0.004	0.665	0.834	0.854	0.005	0.560	0.788	0.825	0.002	0.790	0.947	0.912
	MPA (forest)	**0.008**	**0.245**	**0.584**	**0.617**	**0.009**	**0.090**	**0.562**	**0.545**	**0.004**	**0.642**	**0.887**	**0.846**
	MPA (desert)	0.006	0.452	0.697	0.745	0.007	0.267	0.621	0.662	0.002	0.751	0.927	0.891

introduce noticeable degradation, indicating that visual blending alone poses challenges to camouflage-aware detectors.

When adversarial camouflage is applied, the degradation becomes substantially more pronounced. Both FCA- and transPhyAtt-generated textures further suppress COD performance across all evaluation metrics, while adversarial camouflage optimized by MPA generally leads to the most severe performance degradation. This trend is consistent across different COD architectures and target categories, suggesting that multiprompt optimization disrupts multiple semantic and structural cues simultaneously, making camouflaged targets increasingly difficult to recover even for detectors explicitly designed to handle camouflage.

In addition, civilian vehicles consistently exhibit lower COD performance than armored vehicles across all camouflage settings, reflecting their greater appearance diversity and weaker structural regularities. Among the evaluated architectures, FEDER demonstrates comparatively stronger robustness, followed by UGTR, while MGL exhibits the highest sensitivity to adversarial camouflage. Nevertheless, none of the evaluated COD models is able to reliably recover camouflaged targets under MPA-generated adversarial textures.

Overall, the results in Table XX indicate that existing camouflage-aware detection models remain insufficient to fully mitigate physically realizable adversarial camouflage. These findings highlight the limitations of current COD-based defense strategies and suggest that more comprehensive countermeasures—such as enhanced semantic reasoning, multiview consistency constraints, or multimodal sensing—are necessary to improve the resilience of remote sensing perception systems.

B. Ethical Considerations and Responsible Use

As a physical adversarial attack framework, the proposed MPA method may raise concerns regarding potential misuse in real-world remote sensing scenarios. We emphasize that the primary objective of this work is to expose existing vulnerabilities in OVODs under physically realizable camouflage conditions, thereby supporting the development of more robust, reliable, and secure remote sensing perception systems.

All experiments presented in this article are conducted in controlled simulation environments using publicly available pretrained models and curated datasets. The proposed framework is not intended for malicious deployment, nor does it provide operational guidance for real-world attacks. Instead, MPA is introduced as a threat model to systematically analyze adversarial risks and failure modes of current open-vocabulary detectors, with the goal of informing future defense-oriented research.

We adhere to the principles of responsible disclosure in adversarial robustness research. By evaluating representative COD models as potential defense mechanisms, this work contributes to a broader understanding of the interaction between adversarial attacks and defensive perception models. We encourage future research to build upon these findings to develop effective countermeasures, such as robustness-oriented training strategies, camouflage-aware perception models, and multimodal sensor fusion approaches, while establishing clear ethical boundaries and safe deployment practices for open-vocabulary remote sensing applications.

VI. CONCLUSION

This article presented an MPA framework for generating physically deployable adversarial camouflage to conceal ground vehicles in remote sensing imagery. To the best of our knowledge, this is the first work that systematically studies physically realizable camouflage attacks against OVODs. OVODs introduce a language-conditioned attack surface that is absent in closed-set detection: the same physical target may remain detectable when the textual query changes, even though the image content is unchanged. This cross-prompt transfer uncertainty, coupled with remote sensing viewpoint variation, makes single-prompt camouflage insufficient for robust attack evaluation.

Crucially, beyond the attack formulation itself, we establish a reproducible evaluation foundation for this underexplored

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setting. We construct and release a multiview remote sensing benchmark with high-fidelity 3-D vehicle models, together with a complete data acquisition pipeline that supports scalable extension to new scenes and target categories, allowing the community to reproduce our experiments and construct customized datasets for open-vocabulary physical attack and defense research.

On the methodology side, MPA integrates differentiable rendering to optimize textures under diverse observation configurations, incorporates a controllable color-space constraint to enhance visual plausibility and concealment, and introduces a multiprompt joint attack loss that suppresses prompt-conditioned detector responses across semantically related textual queries. Extensive experiments demonstrate consistently high ASRs across prompts, viewpoints, and observation conditions while maintaining strong visual stealth, and together with the released benchmark and acquisition pipeline, they provide a reproducible testbed for open-vocabulary remote sensing robustness research.

The current evaluation has two scope limitations. First, the benchmark covers two vehicle target types, each represented by a single 3-D model, and does not yet include nonvehicle targets such as ships or aircraft. Second, the victim detector set is limited to GLIP, GroundingDINO, and YOLO-World, which are widely used general-purpose OVODs; evaluating MPA against remote sensing-specific OVODs remains an important next step.

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